

**Vidya Vikas Institute of Engineering and Technology,
Mysore
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VTU Question Papers

“Mechanical Engineering”

IV, VI & VIII Semester

June/July 2023

LIBRARY AND INFORMATION CENTRE

**Vidya Vikas Institute of Engineering and Technology
Bannur road, Alanahalli, Mysore-570028**

INDEX

S/L	Subject Code	Subject Title	Exam Date
1	21BE45	Biology for Engineers	June/July 2023
2	21CIP37/21CIP 47	CONSTITUTION OF INDIA AND PROFESSIONAL ETHICS	June/July 2023
3	21KBK47	Balake Kannada	June/July 2023
4	21KSK47	Samskruthika Kannada	June/July 2023
5	21ME42	Machining Science and Jigs & Fixtures	June/July 2023
6	21ME43	Fluid Mechanics	June/July 2023
7	21ME44	Mechanics and Materials	June/July 2023
8	21UH49	Universal Human Values	June/July 2023
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14	18ME651	Non -Conventional Energy Sources	June/July 2023
15	18ME81	Energy Engineering	June/July 2023
16	18ME824	Automobile Engineering	June/July 2023

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Fourth Semester B.E./B.Tech. Degree Examination, June/July 2023
Biology for Engineers

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	What is a biomolecule? Explain the classification of biomolecule.	07	L2	CO1
	b.	Explain the role of DNA vaccine for rabies and RNA vaccine for COVID-19.	07	L2	CO1
	c.	Write a short note on cellulose based bio-filters.	06	L2	CO1
OR					
Q.2	a.	Explain the DNA finger printing in forensic applications.	07	L2	CO1
	b.	Explain the role of lipids and its application in cleaning agents.	07	L2	CO1
	c.	Write a short note on biosensors and bioplastics.	06	L2	CO1
Module – 2					
Q.3	a.	Explain brain as a CPU system.	07	L3	CO1
	b.	Explain eye as a camera system.	07	L3	CO1
	c.	Write a short note on cardiac pacemaker.	06	L2	CO1
OR					
Q.4	a.	Explain the robotic arms for prosthetics.	07	L3	CO1
	b.	Explain heart as a pump system.	07	L3	CO1
	c.	Write a short note on engineering solutions for Parkinson's disease.	06	L2	CO1
Module – 3					
Q.5	a.	Explain the lungs as a purification system.	07	L3	CO2
	b.	Explain the kidney as filtration system.	07	L3	CO2
	c.	Write a short note on spirometry and ventilator.	06	L2	CO2
OR					
Q.6	a.	Explain muscular and skeletal system as scaffolds.	07	L3	CO2
	b.	Explain bio-engineering solutions for muscular dystrophy and osteoporosis.	07	L3	CO2
	c.	Write a short note on Chronic Obstructive Pulmonary Disease (COPD).	06	L2	CO2
Module – 4					
Q.7	a.	Explain the terms Echolocation Ultrasonography and Sonars.	07	L3	CO3
	b.	Explain the process of Photosynthesis and Photovoltaic cells.	07	L3	CO3
	c.	Write a short note on Bionic leaf, GPS, Bird flight and aircraft.	06	L2	CO3
OR					
Q.8	a.	Explain the terms Lotus leaf effect, Plant Burrs and Super hydrophobic and self-cleaning surfaces.	07	L3	CO3
	b.	Explain the terms Spark skin and Swimsuits, Bullet train using biological concepts.	07	L3	CO3
	c.	Write a short note on Hemoglobin – Based Oxygen Carriers (HBOC's) and Perfluorocarbons (PFC).	06	L2	CO3
Module – 5					
Q.9	a.	Explain the DNA Organic and Biocomputing.	07	L3	CO4
	b.	Explain the Bioimaging and Artificial intelligence for Disease Diagnosis.	07	L3	CO4
	c.	Write a short note on Self healing Bioconcrete.	06	L2	CO4
OR					
Q.10	a.	Explain the importance of Bioimaging.	07	L3	CO4
	b.	Explain Bioremediation and Bio-Mining via microbial surface adsorption.	07	L3	CO4
	c.	Write a short note on Nanomedicines and Bioleaching.	06	L2	CO4

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Question Paper Version : A

Third/Fourth Semester B.E./B.Tech. Degree Examination, June/July 2023
CONSTITUTION OF INDIA AND PROFESSIONAL ETHICS

[Time: 1 hrs.]

[Max. Marks: 50]

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the Fifty questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

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1. When was the Indian constitution enacted and adopted?
a) 26/10/1949 b) 26/11/1949 c) 26/4/1949 d) 26/01/1950
 2. 'We the people of India' are the opening words of the,
a) Preamble of the Indian constitution b) Article 21 of the Indian constitution
c) Fundamental rights d) Directive principles of state policy
 3. Which one of these is the primary source of the Indian constitution?
a) British constitution b) Irish constitution
c) Charter Act of 1833 d) Government of India Act of 1935
 4. The original Indian constitution had :
a) 12 parts, 6 schedule and 320 Articles b) 20 parts, 8 schedule and 380 Articles
c) 12 parts, 8 schedule and 396 Articles d) 12 parts, 10 schedule and 300 Articles
 5. The word 'Sovereign' means that,
a) Supreme in nature b) A country is under dictatorship
c) A country is poor of weak d) A country is strong and powerfull
 6. Directive principles are,
a) Justiciable b) Not practiced at rural levels
c) Non-justiciable d) Associated to the Government worker's
 7. How much time was taken for training the constitution?
a) 1 year, 11 months, 18 days b) 5 year, 11 months, 18 days
c) 2 year, 11 months, 18 days d) 3 year, 11 months, 18 days

8. India is a Sovereign, socialist, secular, democratic and republic in the Indian constitution this expression occurs in,
a) Citizenship
b) Preamble
c) Fundamental rights
d) Directive principles
9. Who among the following is the supreme commander of the Armed forces?
a) Air Chief Marshal
b) Prime Minister
c) Defense Minister
d) President
10. The 91st Amendment Act (2003) is associated with,
a) Size of the council of ministers
b) Primary education
c) Fundamental Duty
d) Powers of the President
11. How many members were nominated to the parliament by the president of India?
a) 14 members
b) 12 members
c) 2 members
d) 6 members
12. Who among the following distribute portfolios for the council of minister,
a) President
b) Vice president
c) Prime Minister
d) Speaker of Lok Sabha
13. The chief justice and other judges of the supreme court hold office till they complete,
a) Sixty years
b) Sixty five years
c) Sixty two years
d) Seventy years
14. The council of ministers are responsible to the,
a) Rajya Sabha
b) Vidhan Parishad
c) Lok Sabha
d) Supreme court
15. The Vice-President of India is elected by the,
a) Judges of the supreme court
b) President
c) Prime Minister
d) Members of parliament
16. Who can issue ordinance when the parliament is not in session:
a) President
b) High court judges
c) Home minister
d) Finance minister
17. In case of the violation of the Fundamental Rights we may approach the,
a) Civil Courts
b) Supreme Court
c) High Court
d) Both (a) and (b)
18. Which of the following equalities is/are included in the Right to Equality?
a) Equality before law
b) Equal protection of law
c) Equal opportunities in the public employment
d) All of these.
19. Prohibition of trafficking in human beings and forced labour comes under which of the following fundamental right?
a) Right to freedom
b) Right against exploitation
c) Cultural & Educational Right
d) Right to equality.
20. There is no provision in the constitution for the impeachment of the,
a) President
b) Vice President
c) Governor
d) Supreme court Judges
21. The tenure of the Council of state is,
a) Not subject to dissolution
b) 2 years
c) 5 years
d) 4 years
22. When elections are held in one or a few constituencies due to death or resignation of candidates, it is called as _____.
a) General election
b) Primary election
c) By election
d) Midterm election

23. Fundamental Rights are borrowed from the constitution of,
a) UK b) USA c) Germany d) Ireland
24. What is the minimum age to become Judges of Supreme Court of India?
a) 25 years b) 30 years c) 35 years d) None of these
25. The Indian constitution gives the power of amending the constitution to,
a) The people of India b) The president
c) The Parliament d) Supreme Court of India
26. Right to Education (RTE) was introduced in _____ Amendment,
a) 86th b) 42nd c) 44th d) 61st
27. How many types of writs can be issued by the Supreme Court for the protection of Fundamental Rights?
a) Four b) Five c) One d) Six
28. Who presides over the sessions of Rajya Sabha?
a) Speaker b) Home minister c) Vice-president d) President
29. Who appoints the Vice-Chancellors of the state universities?
a) Education minister b) District commissioner c) Chief minister d) Governor
30. Election commission is a _____ body and the term of election commission is _____ years or _____ years of age whichever is earlier.
a) Uni-member, 4 years or 62 years b) Multi-member, 6 years or 65 years
c) Constitutional body, 5 years or 60 years d) None of these
31. The member to be nominated by the President for the council of states are from,
a) Literature b) Science c) Sports d) All of these
32. Which of the following Pairs is not property matched ,
a) 44th Amendment-citizenship act b) 52nd Amendment-Anti Defection Law
c) 42nd Amendment-Fundamental duties d) 73rd Amendment-Local self Government
33. The speaker of Lok Sabha,
a) is appointed by the President b) is nominated by the Vice-President
c) is chosen by the members of Lok Sabha d) is elected by the members of parliament
34. Financial Emergency has been imposed in India,
a) Once b) Never c) Twice d) Thrice
35. Respect for the National Flag and the National Anthem is,
a) a Fundamental right b) a Fundamental Duty
c) a Directive principle d) an ordinary duty
36. A non-member of the state legislature can be the minister for a period not exceeding,
a) Six month b) One year c) Six weeks d) Three months
37. Engineering ethics is a,
a) developing ethics b) Preventive ethics
c) natural ethics d) Scientifically developed ethics

38. Risk estimation can be done by using,
a) Cooking b) Trimming c) Event tree d) Both (a) and (b)
39. The Patent holder does not allow others to use patented information for _____ years from the date of filing.
a) 25 b) 30 c) 50 d) 20
40. The use of intellectual property of others without their permission or credit is referred to as,
a) Cooking b) Plagiarism c) Patents d) Formulae
41. Which of the following is not the concept of responsibilities?
a) Minimalist b) Reasonable care c) Utilitarianism d) Good works
42. Lying means,
a) Intentionally conveying false information to others b) Fabrication
c) Plagiarism d) All of these
43. The three types of Justice referred in our preamble are :
a) Social, Economic and Religious b) Social, Economic and Natural
c) Social, Economic and International d) Social, Economic and Political
44. An arrested person must be produced before a magistrate within _____ hours of arrest.
a) 12 b) 24 c) 36 d) 48
45. Election commission conducts the election as per which act?
a) Parliament act b) People's representative act of 1982
c) Code of conduct act d) State representative act
46. When the office of the president, falls vacant, the same must be filled up with in?
a) 3 months b) 6 months c) 1 year d) 9 months
47. Who among the following are not entitled to form Union or Association,
a) Police b) Teachers c) Workers d) Doctors
48. The MLA's of various state legislative assemblies are varying between,
a) 40 to 450 b) 50 - 500 c) 28 - 12 d) 60 - 500
49. A bill cannot become an act of parliament, unless and until _____?
a) it is passed by Lok Sabha b) it is passed by Rajya Sabha
c) it gets assent from President d) it gets approved by Supreme Court
50. Who hoisted the National Flag during 74th Republic day function in New Delhi?
a) Prime Minister b) President
c) Vice-President d) Chief justice of India

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Question Paper Version : B

Third/Fourth Semester B.E./B.Tech. Degree Examination, June/July 2023
CONSTITUTION OF INDIA AND PROFESSIONAL ETHICS

[Time: 1 hrs.]

[Max. Marks: 50]

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Question Paper Version : C**Third/Fourth Semester B.E. /B.Tech. Degree Examination, June/July 2023****CONSTITUTION OF INDIA AND PROFESSIONAL ETHICS**

[Time: 1 hrs.]

[Max. Marks: 50]

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a) Judges of the supreme court b) President
c) Prime Minister d) Members of parliament
46. Who can issue ordinance when the parliament is not in session:
a) President b) High court judges c) Home minister d) Finance minister
47. In case of the violation of the Fundamental Rights we may approach the,
a) Civil Courts b) Supreme Court c) High Court d) Both (a) and (b)
48. Which of the following equalities is/are included in the Right to Equality?
a) Equality before law b) Equal protection of law
c) Equal opportunities in the public employment d) All of these.
49. Prohibition of trafficking in human beings and forced labour comes under which of the following fundamental right?
a) Right to freedom b) Right against exploitation
c) Cultural & Educational Right d) Right to equality.
50. There is no provision in the constitution for the impeachment of the,
a) President b) Vice President c) Governor d) Supreme court Judges

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Question Paper Version : D

Third/Fourth Semester B.E./B.Tech. Degree Examination, June/July 2023
CONSTITUTION OF INDIA AND PROFESSIONAL ETHICS

[Time: 1 hrs.]

[Max. Marks: 50]

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the Fifty questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

-
1. The tenure of the Council of state is,
a) Not subject to dissolution b) 2 years c) 5 years d) 4 years
 2. When elections are held in one or a few constituencies due to death or resignation of candidates, it is called as _____.
a) General election b) Primary election
c) By election d) Midterm election
 3. Fundamental Rights are borrowed from the constitution of,
a) UK b) USA c) Germany d) Ireland
 4. What is the minimum age to become Judges of Supreme Court of India?
a) 25 years b) 30 years c) 35 years d) None of these
 5. The Indian constitution gives the power of amending the constitution to,
a) The people of India b) The president
c) The Parliament d) Supreme Court of India
 6. Right to Education (RTE) was introduced in _____ Amendment,
a) 86th b) 42nd c) 44th d) 61st
 7. How many types of writs can be issued by the Supreme Court for the protection of Fundamental Rights?
a) Four b) Five c) One d) Six
 8. Who presides over the sessions of Rajya Sabha?
a) Speaker b) Home minister c) Vice-president d) President

9. Who appoints the Vice-Chancellors of the state universities?
a) Education minister b) District commissioner c) Chief minister d) Governor
10. Election commission is a _____ body and the term of election commission is _____ years or _____ years of age whichever is earlier.
a) Uni-member, 4 years or 62 years b) Multi-member, 6 years or 65 years
c) Constitutional body, 5 years or 60 years d) None of these
11. Which of the following is not the concept of responsibilities?
a) Minimalist b) Reasonable care c) Utilitarianism d) Good works
12. Lying means,
a) Intentionally conveying false information to others b) Fabrication
c) Plagiarism d) All of these
13. The three types of Justice referred in our preamble are :
a) Social, Economic and Religious b) Social, Economic and Natural
c) Social, Economic and International d) Social, Economic and Political
14. An arrested person must be produced before a magistrate within _____ hours of arrest.
a) 12 b) 24 c) 36 d) 48
15. Election commission conducts the election as per which act?
a) Parliament act b) People's representative act of 1982
c) Code of conduct act d) State representative act
16. When the office of the president, falls vacant, the same must be filled up with in?
a) 3 months b) 6 months c) 1 year d) 9 months
17. Who among the following are not entitled to form Union or Association,
a) Police b) Teachers c) Workers d) Doctors
18. The MLA's of various state legislative assemblies are varying between,
a) 40 to 450 b) 50 - 500 c) 28 - 12 d) 60 - 500
19. A bill cannot become an act of parliament, unless and until _____?
a) it is passed by Lok Sabha b) it is passed by Rajya Sabha
c) it gets assent from President d) it gets approved by Supreme Court
20. Who hoisted the National Flag during 74th Republic day function in New Delhi?
a) Prime Minister b) President
c) Vice-President d) Chief justice of India
21. The member to be nominated by the President for the council of states are from,
a) Literature b) Science c) Sports d) All of these
22. Which of the following Pairs is not properly matched ,
a) 44th Amendment-citizenship act b) 52nd Amendment-Anti Defection Law
c) 42nd Amendment-Fundamental duties d) 73rd Amendment-Local self Government
23. The speaker of Lok Sabha,
a) is appointed by the President b) is nominated by the Vice-President
c) is chosen by the members of Lok Sabha d) is elected by the members of parliament

24. Financial Emergency has been imposed in India,
a) Once b) Never c) Twice d) Thrice
25. Respect for the National Flag and the National Anthem is,
a) a Fundamental right b) a Fundamental Duty
c) a Directive principle d) an ordinary duty
26. A non-member of the state legislature can be the minister for a period not exceeding,
a) Six month b) One year c) Six weeks d) Three months
27. Engineering ethics is a,
a) developing ethics b) Preventive ethics
c) natural ethics d) Scientifically developed ethics
28. Risk estimation can be done by using,
a) Cooking b) Trimming c) Event tree d) Both (a) and (b)
29. The Patent holder does not allow others to use patented information for _____ years from the date of filing.
a) 25 b) 30 c) 50 d) 20
30. The use of intellectual property of others without their permission or credit is referred to as,
a) Cooking b) Plagiarism c) Patents d) Formulae
31. How many members were nominated to the parliament by the president of India?
a) 14 members b) 12 members c) 2 members d) 6 members
32. Who among the following distribute portfolios for the council of minister,
a) President b) Vice president c) Prime Minister d) Speaker of Lok Sabha
33. The chief justice and other judges of the supreme court hold office till they complete,
a) Sixty years b) Sixty five years c) Sixty two years d) Seventy years
34. The council of ministers are responsible to the,
a) Rajya Sabha b) Vidhan Parishad c) Lok Sabha d) Supreme court
35. The Vice-President of India is elected by the,
a) Judges of the supreme court b) President
c) Prime Minister d) Members of parliament
36. Who can issue ordinance when the parliament is not in session:
a) President b) High court judges c) Home minister d) Finance minister
37. In case of the violation of the Fundamental Rights we may approach the,
a) Civil Courts b) Supreme Court c) High Court d) Both (a) and (b)
38. Which of the following equalities is/are included in the Right to Equality?
a) Equality before law b) Equal protection of law
c) Equal opportunities in the public employment d) All of these.
39. Prohibition of trafficking in human beings and forced labour comes under which of the following fundamental right?
a) Right to freedom b) Right against exploitation
c) Cultural & Educational Right d) Right to equality.

40. There is no provision in the constitution for the impeachment of the,
a) President b) Vice President c) Governor d) Supreme court Judges
41. When was the Indian constitution enacted and adopted?
a) 26/10/1949 b) 26/11/1949 c) 26/4/1949 d) 26/01/1950
42. 'We the people of India' are the opening words of the,
a) Preamble of the Indian constitution b) Article 21 of the Indian constitution
c) Fundamental rights d) Directive principles of state policy
43. Which one of these is the primary source of the Indian constitution?
d) British constitution b) Irish constitution
c) Charter Act of 1833 d) Government of India Act of 1935
44. The original Indian constitution had :
a) 12 parts, 6 schedule and 320 Articles b) 20 parts, 8 schedule and 380 Articles
c) 12 parts, 8 schedule and 396 Articles d) 12 parts, 10 schedule and 300 Articles
45. The word 'Sovereign' means that,
a) Supreme in nature b) A country is under dictatorship
c) A country is poor of weak d) A country is strong and powerfull
46. Directive principles are,
a) Justiciable b) Not practiced at rural levels
c) Non-justiciable d) Associated to the Government worker's
47. How much time was taken for training the constitution?
a) 1 year, 11 months, 18 days b) 5 year, 11 months, 18 days
c) 2 year, 11 months, 18 days d) 3 year, 11 months, 18 days
48. India is a Sovereign, socialist, selular, democratic and republic in the Indian constitution this expression occurs in,
a) Citizenship b) Preamble
c) Fundamental rights d) Directive principles
49. Who among the following is the supreme commander of the Armed forces?
a) Air Chief Marshal b) Prime Minister
c) Defense Minister d) President
50. The 91st Amendment Act (2003) is associated with,
a) Size of the council of ministers b) Primary education
c) Fundamental Duty d) Powers of the President

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Question Paper Version : A

Fourth Semester B.E./B.Tech. Degree Examination, June/July 2023
Balake Kannada

[Time: 1 hrs.]

[Max. Marks: 50]

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the Fifty questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

Note : Fill in the blanks with suitable words given :

1. Nimma heSarU _____?
a) Yaaru b) eenu c) elli d) YaaVaga
2. Ee Hannina _____estu?
a) Banna b) Ruchi c) Bele d) Tuka
3. _____hudugi
a) Avalu b) Avaru c) Avanu d) Ivaru
4. Nimma KaaleJu _____?
a) Ellige b) Ellide c) Elli d) Ellinda
5. Ravi nanna _____ (Friend)
a) Tamma b) Anna c) Snehita d) ShikShaka

Note: Translate English words to Kannada :

6. Laboratory _____.
a) PraYogalaya b) Granthalaya c) Taragati d) Adhyayana
7. Rented house _____.
a) Angadi b) Ruumu c) Swantha Mane d) Badige mane
8. Heritage _____.
a) Samskruthi b) Habba c) parampare d) UthSaVa

9. People _____
a) Vidyarthi b) Jana c) Jaana d) Mara
10. Water _____.
a) HaalU b) Male c) Kere d) NeerU
11. Child _____.
a) Magalu b) Huduga c) Magu d) Mahile
12. Good night _____.
a) Vandane b) Munjane c) shubharaatri d) Shubhadina

Note: Substitute the word from the following each sentence in appropriate places:

13. Raja : Rajanaddu :: Rani : _____
a) Raniyaddu b) Raviyaddu c) Raniyaru d) Raja Rani
14. Mane : Maneyalli :: uuru : _____
a) eelli b) Nagardalli c) uurinalli d) Halliyalli
15. HoSa : HoSadu :: HaLeya : _____
a) Doddadu b) HaLeyadu c) Avaladu d) Maneyadu
16. oota : ootakke :: Aata : _____
a) patakke b) AataVadu c) Aatadalli d) Aatakke
17. Mane : Manege :: KaaleJu : _____
a) Kaalige b) KaaleJalli c) Kaalejige d) KaaLeJuyelli
18. Mara : Maragalu :: Hannu : _____
a) Hannige b) Hannalli c) Henu d) Hannugalu

Note: Write the English word for the given Kannada word:

19. PradhyaPaKa _____
a) Teacher b) Warden c) Professor d) Principal
20. Pashchima _____
a) East b) West c) South d) North
21. Varsha _____
a) Month b) Week c) Year d) Day
22. RaSaayanaShastra _____
a) Chemistry b) Physics c) Botony d) Mathematics
23. HaSiru _____
a) Red b) Blue c) Brown d) Green

24. Mara _____
a) Plant b) Earth c) Tree d) Soil
25. Granthalaya _____
a) Laboratory b) Library c) Book d) Classroom

Note: Translate following sentences into English:

26. Niivu Yarru?
a) How are you? b) Who is he?
c) Who are you? d) Where are you?
27. Ninna KaaleJu Ellide?
a) What is your name? b) Where is your house?
c) Where is your college? d) What do you do?
28. Avala heSaru Kamala.
a) Her name is Kamala. b) Kamala where are you?
c) His name is Kamala. d) Kamala good girl.
29. Idu nanna mane.
a) Where is your house? b) Which is your place?
c) This is my house. d) What is your house name?
30. Bengaluru dodda Nagara.
a) My place is Bangalore. b) Bangalore is a green city.
c) Bangalore is a city. d) Bangalore is a big city.

Note: Choose the appropriate answer:

31. Country _____.
a) Rajya b) DeSha c) prapancha d) VideSha
32. Choose the mathematical '+' symbol in Kannada.
a) Kale b) Koodu c) Gunisu d) BaagiSu
33. Daughter _____.
a) Maga b) Akka c) Hudugi d) Magalu
34. Samskruthi _____.
a) Festival b) Heritage c) Culture d) All of these
35. Aarogya _____.
a) Wealth b) Health c) Patient d) Hospital
36. Them _____.
a) Avaru b) NaaVu c) Avanu d) elli

37. HoSadu _____ (Identify the Kannada opposite word)
 a) Mele b) Kadime c) Belaku d) Haleyadu
38. Young sister _____.
 a) Tamma b) Tande c) Appa d) Tangi
39. Drama _____.
 a) Kate b) Paata c) Naataka d) Banna
40. Rent _____.
 a) Mane b) Bele c) Angadi d) Baadige

Note : Fill in the blanks with appropriate Kannada Word:

41. Naale _____uurige hogonaVaa? (we)
 a) Naavu b) adu c) Yaaru d) AVarU
42. Nimma _____Yaavudu? (Village)
 a) Mane b) pattana c) uuru d) byagu
43. Naanu _____bengulurige hogue? (Tomorrow)
 a) Indu b) Naadiddu c) Bega d) Naale
44. Nimma Kaaleju _____? (which)
 a) Yaake b) elli c) YaaVudu d) Hege
45. Nimma _____enu? (Name)
 a) uuru b) Mane c) Kaaleju d) Hesaru
46. Adu Namma _____ (house)
 a) Shaale b) uuru c) Pennu d) Mane

Note : Translate following Kannada sentences into English

47. Adu Enu?
 a) Who is this? b) Where is this? c) What is that? d) What is this?
48. Krishna Yaaru?
 a) Who is Krishna? b) Where is Krishna? c) Who is he? d) Who are you?
49. Niivu Yaaru?
 a) Who is she? b) Who are you? c) Who is this? d) How is?
50. AVaru Namma Tande.
 a) He is my son. b) She is my daughter.
 c) This is my place. d) He is my Father.

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Question Paper Version : B

Fourth Semester B.E./B.Tech. Degree Examination, June/July 2023
Balake Kannada

[Time: 1 hrs.]

[Max. Marks: 50]

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the Fifty questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

Note : Translate following Kannada sentences into English

1. Adu Enu?
a) Who is this? b) Where is this? c) What is that? d) What is this?
2. Krishna Yaaru?
a) Who is Krishna? b) Where is Krishna? c) Who is he? d) Who are you?
3. Niivu Yaaru?
a) Who is she? b) Who are you? c) Who is this? d) How is?
4. AVaru Namma Tande.
a) He is my son. b) She is my daughter.
c) This is my place. d) He is my Father.

Note : Fill in the blanks with appropriate Kannada Word:

5. Naale _____uurige hogonaVaa? (we)
a) Naavu b) adu c) Yaaru d) AVarU
6. Nimma _____Yaavudu? (Village)
a) Mane b) pattana c) uuru d) byagu
7. Naanu _____bengulurige hogue? (Tomorrow)
a) Indu b) Naadiddu c) Bega d) Naale
8. Nimma Kaaleju _____? (which)
a) Yaake b) elli c) YaaVudu d) Hege

9. Nimma _____enu? (Name)
a) uuru b) Mane c) Kaaleju d) Hesaru
10. Adu Namma _____ (house)
a) Shaale b) uuru c) Pennu d) Mane

Note: Choose the appropriate answer:

11. Country _____.
a) Rajya b) DeSha c) prapancha d) VideSha
12. Choose the mathematical '+' symbol in Kannada.
a) Kale b) Koodu c) Gunisu d) BaagiSu
13. Daughter _____.
a) Maga b) Akka c) Hudugi d) Magalu
14. Samskruthi _____.
a) Festival b) Heritage c) Culture d) All of these
15. Aarogya _____.
a) Wealth b) Health c) Patient d) Hospital
16. Them _____.
a) Avaru b) NaaVu c) Avanu d) elli
17. HoSadu _____(Identify the Kannada opposite word)
a) Mele b) Kadime c) Belaku d) Haleyadu
18. Young sister _____.
a) Tamma b) Tande c) Appa d) Tangi
19. Drama _____.
a) Kate b) Paata c) Naataka d) Banna
20. Rent _____.
a) Mane b) Bele c) Angadi d) Baadige

Note: Translate following sentences into English:

21. Niivu Yarru?
a) How are you? b) Who is he?
c) Who are you? d) Where are you?
22. Ninna KaaleJu Ellide?
a) What is your name? b) Where is your house?
c) Where is your college? d) What do you do?
23. Avala heSaru Kamala.
a) Her name is Kamala. b) Kamala where are you?
c) His name is Kamala. d) Kamala good girl.

24. Idu nanna mane.
 a) Where is your house? b) Which is your place?
 c) This is my house. d) What is your house name?
25. Bengaluru dodda Nagara.
 a) My place is Bangalore. b) Bangalore is a green city.
 c) Bangalore is a city. d) Bangalore is a big city.

Note: Write the English word for the given Kannada word:

26. PradhyaPaKa _____
 a) Teacher b) Warden c) Professor d) Principal
27. Pashchima _____
 a) East b) West c) South d) North
28. Varsha _____
 a) Month b) Week c) Year d) Day
29. RaSaayanaShastra _____
 a) Chemistry b) Physics c) Botony d) Mathematics
30. HaSiru _____
 a) Red b) Blue c) Brown d) Green
31. Mara _____
 a) Plant b) Earth c) Tree d) Soil
32. Granthalaya _____
 a) Laboratory b) Library c) Book d) Classroom

Note: Substitute the word from the following each sentence in appropriate places:

33. Raja : Rajanaddu :: Rani : _____
 a) Raniyaddu b) Raviyaddu c) Raniyaru d) Raja Rani
34. Mane : Maneyalli :: uuru : _____
 a) eelli b) Nagardalli c) uurinalli d) Halliyalli
35. HoSa : HoSadu :: HaLeya : _____
 a) Doddadu b) HaLeyadu c) Avaladu d) Maneyadu
36. oota : ootakke :: Aata : _____
 a) patakke b) AataVadu c) Aatadalli d) Aatakke
37. Mane : Manege :: KaaleJu : _____
 a) Kaalige b) KaaleJalli c) Kaalejige d) KaaLeJuyelli
38. Mara : Maragalu :: Hannu : _____
 a) Hannige b) Hannalli c) Hennu d) Hannugalu

Note: Translate English words to Kannada :

39. Laboratory _____.
a) PraYogalaya b) Granthalaya c) Taragati d) Adhyayana
40. Rented house _____.
a) Angadi b) Ruumu c) Swantha Mane d) Badige mane
41. Heritage _____.
a) Samskruthi b) Habba c) parampare d) UthSaVa
42. People _____.
a) Vidyarthi b) Jana c) Jaana d) Mara
43. Water _____.
a) HaalU b) Male c) Kere d) NeerU
44. Child _____.
a) Magalu b) Huduga c) Magu d) Mahile
45. Good night _____.
a) Vandane b) Munjane c) shubharaatri d) Shubhadina

Note : Fill in the blanks with suitable words given :

46. Nimma heSarU _____.
a) Yaaru b) eenu c) elli d) YaaVaga
47. Ee Hannina _____estu?
a) Banna b) Ruchi c) Bele d) Tuka
48. _____hudugi
a) Avalu b) Avaru c) Avanu d) Ivaru
49. Nimma KaaleJu _____.
a) Ellige b) Ellide c) Elli d) Ellinda
50. Ravi nanna _____(Friend)
a) Tamma b) Anna c) Snehita d) ShikShaka

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Question Paper Version : C

Fourth Semester B.E./B.Tech. Degree Examination, June/July 2023
Balake Kannada

[Time: 1 hrs.]

[Max. Marks: 50]

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the Fifty questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

Note: Translate English words to Kannada :

1. Laboratory _____.
a) PraYogalaya b) Granthalaya c) Taragati d) Adhyayana
2. Rented house _____.
a) Angadi b) Ruumu c) Swantha Mane d) Badige mane
3. Heritage _____.
a) Samskruthi b) Habba c) parampare d) UthSaVa
4. People _____.
a) Vidyarthi b) Jana c) Jaana d) Mara
5. Water _____.
a) HaalU b) Male c) Kere d) NeerU
6. Child _____.
a) Magalu b) Huduga c) Magu d) Mahile
7. Good night _____.
a) Vandane b) Munjane c) shubharaatri d) Shubhadina

Note: Write the English word for the given Kannada word:

8. PradhyaPaKa _____.
a) Teacher b) Warden c) Professor d) Principal
9. Pashchima _____.
a) East b) West c) South d) North

10. Varsha _____
a) Month b) Week c) Year d) Day
11. RaSaayanaShastra _____
a) Chemistry b) Physics c) Botony d) Mathematics
12. HaSiru _____
a) Red b) Blue c) Brown d) Green
13. Mara _____
a) Plant b) Earth c) Tree d) Soil
14. Granthalaya _____
a) Laboratory b) Library c) Book d) Classroom

Note : Translate following Kannada sentences into English

15. Adu Enu?
a) Who is this? b) Where is this? c) What is that? d) What is this?
16. Krishna Yaaru?
a) Who is Krishna? b) Where is Krishna? c) Who is he? d) Who are you?
17. Niivu Yaaru?
a) Who is she? b) Who are you? c) Who is this? d) How is?
18. AVaru Namma Tande.
a) He is my son. b) She is my daughter.
c) This is my place. d) He is my Father.

Note: Choose the appropriate answer:

19. Country _____.
a) Rajya b) DeSha c) prapancha d) VideSha
20. Choose the mathematical '+' symbol in Kannada.
a) Kale b) Koodu c) Gunisu d) BaagiSu
21. Daughter _____.
a) Maga b) Akka c) Hudugi d) Magalu
22. Samskruthi _____.
a) Festival b) Heritage c) Culture d) All of these
23. Aarogya _____.
a) Wealth b) Health c) Patient d) Hospital
24. Them _____.
a) Avaru b) NaaVu c) Avanu d) elli

25. HoSadu _____ (Identify the Kannada opposite word)
 a) Mele b) Kadime c) Belaku d) Haleyadu
26. Young sister _____
 a) Tamma b) Tande c) Appa d) Tangi
27. Drama _____
 a) Kate b) Paata c) Naataka d) Banna
28. Rent _____
 a) Mane b) Bele c) Angadi d) Baadige

Note : Fill in the blanks with suitable words given :

29. Nimma heSarU _____ ?
 a) Yaaru b) eenu c) elli d) YaaVaga
30. Ee Hannina _____ estu?
 a) Banna b) Ruchi c) Bele d) Tuka
31. _____ hudugi
 a) Avalu b) Avaru c) Avanu d) Ivaru
32. Nimma KaaleJu _____ ?
 a) Ellige b) Ellide c) Elli d) Ellinda
33. Ravi nanna _____ (Friend)
 a) Tamma b) Anna c) Snehita d) ShikShaka

Note : Fill in the blanks with appropriate Kannada Word:

34. Naale _____ uurige hogonaVaa? (we)
 a) Naavu b) adu c) Yaaru d) AVarU
35. Nimma _____ Yaavudu? (Village)
 a) Mane b) pattana c) uuru d) byagu
36. Naanu _____ bengulurige hogueve? (Tomorrow)
 a) Indu b) Naadiddu c) Bega d) Naale
37. Nimma Kaaleju _____ ? (which)
 a) Yaake b) elli c) YaaVudu d) Hege
38. Nimma _____ enu? (Name)
 a) uuru b) Mane c) Kaaleju d) Hesaru
39. Adu Namma _____ (house)
 a) Shaale b) uuru c) Pennu d) Mane

Note: Translate following sentences into English:

40. Niivu Yarru?
a) How are you? b) Who is he?
c) Who are you? d) Where are you?
41. Ninna KaaleJu Ellide?
a) What is your name? b) Where is your house?
c) Where is your college? d) What do you do?
42. Avala heSaru Kamala.
a) Her name is Kamala. b) Kamala where are you?
c) His name is Kamala. d) Kamala good girl.
43. Idu nanna mane.
a) Where is your house? b) Which is your place?
c) This is my house. d) What is your house name?
44. Bengaluru dodda Nagara.
a) My place is Bangalore. b) Bangalore is a green city.
c) Bangalore is a city. d) Bangalore is a big city.

Note: Substitute the word from the following each sentence in appropriate places:

45. Raja : Rajanaddu :: Rani : _____
a) Raniyaddu b) Raviyaddu c) Raniyaru d) Raja Rani
46. Mane : Maneyalli :: uuru : _____
a) eelli b) Nagardalli c) uurinalli d) Halliyalli
47. HoSa : HoSadu :: HaLeya : _____
a) Doddadu b) HaLeyadu c) Avaladu d) Maneyadu
48. oota : ootakke :: Aata : _____
a) patakke b) AataVadu c) Aatadalli d) Aatakke
49. Mane : Manege :: KaaleJu : _____
a) Kaalige b) KaaleJalli c) Kaalejige d) KaaLeJuyelli
50. Mara : Maragalu :: Hannu : _____
a) Hannige b) Hannalli c) Hennu d) Hannugalu

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Fourth Semester B.E./B.Tech. Degree Examination, June/July 2023
Balake Kannada

[Time: 1 hrs.]

[Max. Marks: 50]

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the Fifty questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

Note: Translate following sentences into English:

1. Niivu Yarru?
 - a) How are you?
 - b) Who is he?
 - c) Who are you?
 - d) Where are you?
2. Ninna KaaleJu Ellide?
 - a) What is your name?
 - b) Where is your house?
 - c) Where is your college?
 - d) What do you do?
3. Avala heSaru Kamala.
 - a) Her name is Kamala.
 - b) Kamala where are you?
 - c) His name is Kamala.
 - d) Kamala good girl.
4. Idu nanna mane.
 - a) Where is your house?
 - b) Which is your place?
 - c) This is my house.
 - d) What is your house name?
5. Bengaluru dodda Nagara.
 - a) My place is Bangalore.
 - b) Bangalore is a green city.
 - c) Bangalore is a city.
 - d) Bangalore is a big city.

Note: Substitute the word from the following each sentence in appropriate places:

6. Raja : Rajanaddu :: Rani : _____
a) Raniyaddu b) Raviyaddu c) Raniyaru d) Raja Rani
7. Mane : Maneyalli :: uuru : _____
a) eelli b) Nagardalli c) uurinalli d) Halliyalli

8. HoSa : HoSadu :: HaLeya : _____
 a) Doddadu b) HaLeyadu c) Avaladu d) Maneyadu
9. oota : ootakke :: Aata : _____
 a) patakke b) AataVadu c) Aatadalli d) Aatakke
10. Mane : Manege :: KaaleJu : _____
 a) Kaalige b) KaaleJalli c) Kaalejige d) KaaLeJuyelli
11. Mara : Maragalu :: Hannu : _____
 a) Hannige b) Hannalli c) Hennu d) Hannugalu

Note : Fill in the blanks with suitable words given :

12. Nimma heSarU _____?
 a) Yaaru b) eenu c) elli d) YaaVaga
13. Ee Hannina _____estu?
 a) Banna b) Ruchi c) Bele d) Tuka
14. _____hudugi
 a) Avalu b) Avaru c) Avanu d) Ivaru
15. Nimma KaaleJu _____?
 a) Ellige b) Ellide c) Elli d) Ellinda
16. Ravi nanna _____(Friend)
 a) Tamma b) Anna c) Snehita d) ShikShaka

Note : Fill in the blanks with appropriate Kannada Word:

17. Naale _____uurige hogonaVaa? (we)
 a) Naavu b) adu c) Yaaru d) AVarU
18. Nimma _____Yaavudu? (Village)
 a) Mane b) pattana c) uuru d) byagu
19. Naanu _____bengulurige hogue? (Tomorrow)
 a) Indu b) Naadiddu c) Bega d) Naale
20. Nimma Kaaleju _____? (which)
 a) Yaake b) elli c) YaaVudu d) Hege
21. Nimma _____enu? (Name)
 a) uuru b) Mane c) Kaaleju d) Hesaru
22. Adu Namma _____(house)
 a) Shaale b) uuru c) Pennu d) Mane

Note: Choose the appropriate answer:

23. Country _____.
a) Rajya b) DeSha c) prapancha d) VideSha
24. Choose the mathematical '+' symbol in Kannada.
a) Kale b) Koodu c) Gunisu d) BaagiSu
25. Daughter _____.
a) Maga b) Akka c) Hudugi d) Magalu
26. Samskruthi _____.
a) Festival b) Heritage c) Culture d) All of these
27. Aarogya _____.
a) Wealth b) Health c) Patient d) Hospital
28. Them _____.
a) Avaru b) NaaVu c) Avanu d) elli
29. HoSadu _____ (Identify the Kannada opposite word)
a) Mele b) Kadime c) Belaku d) Haleyadu
30. Young sister _____.
a) Tamma b) Tande c) Appa d) Tangi
31. Drama _____.
a) Kate b) Paata c) Naataka d) Banna
32. Rent _____.
a) Mane b) Bele c) Angadi d) Baadige

Note: Translate English words to Kannada :

33. Laboratory _____.
a) PraYogalaya b) Granthalaya c) Taragati d) Adhyayana
34. Rented house _____.
a) Angadi b) Ruumu c) Swantha Mane d) Badige mane
35. Heritage _____.
a) Samskruthi b) Habba c) parampare d) UthSaVa
36. People _____.
a) Vidyarthi b) Jana c) Jaana d) Mara
37. Water _____.
a) HaalU b) Male c) Kere d) NeerU

38. Child _____.
a) Magalu b) Huduga c) Magu d) Mahile
39. Good night _____.
a) Vandane b) Munjane c) shubharaatri d) Shubhadina

Note : Translate following Kannada sentences into English

40. Adu Enu?
a) Who is this? b) Where is this? c) What is that? d) What is this?
41. Krishna Yaaru?
a) Who is Krishna? b) Where is Krishna? c) Who is he? d) Who are you?
42. Niivu Yaaru?
a) Who is she? b) Who are you? c) Who is this? d) How is?
43. AVaru Namma Tande.
a) He is my son. b) She is my daughter.
c) This is my place. d) He is my Father.

Note: Write the English word for the given Kannada word:

44. PradhyaPaKa _____.
a) Teacher b) Warden c) Professor d) Principal
45. Pashchima _____.
a) East b) West c) South d) North
46. Varsha _____.
a) Month b) Week c) Year d) Day
47. RaSaayanaShastra _____.
a) Chemistry b) Physics c) Botony d) Mathematics
48. HaSiru _____.
a) Red b) Blue c) Brown d) Green
49. Mara _____.
a) Plant b) Earth c) Tree d) Soil
50. Granthalaya _____.
a) Laboratory b) Library c) Book d) Classroom

* * * * *

[illegible]

Fourth Semester B.E/B.Tech Degree Examination, June/July 2023

ಉಂಸಾಹಕ ಕನಡ - Samskruthika Kannada

Time: 1 hrs.]

[Max. Marks: 50]

ÀsZÀ£*nÀ¼Àc

1. CW! BO YÁ±*BnÀ½nÀS wvÀÛ&1&. YÁ, & YÁ, ±*Bn* wozÀc CoaÁ.
2. c.co.Dg@ wvÀÛgÀ YÁ&a*oÀceè oÀcc.c, @.c&@, Ào6*, °ÁnÀs YÁ±*B YÁ&a*oÀc ±*Á-°oÀc&ÀcB CozÀg* A, B, C CxÀªÁ D oÀc&ÀcB vÁ!, wèzÀov* a&&ÁoÀcªÁV nÀcgÀc&, ÀcªÁÀzÀc CªÁ, %7oÀc vªÁªÁÁ&ÁVgÀcvÀÛz*.
3. c.co.Dg@ wvÀÛgÀ YÁ&a*oÀceè enÀ&YÀr¹gÀcªÁ, ÀÛ¼ÁzÀeè ªÁ&7ªÁcÁ&zª* °Án*oªcÁ ele CxÀªÁ ªÁ&7ªÁÁr zÀªÁªÁ>&oÀceè oÀcÁªªÁzª*ÁªÀ vÁ, ÀezÁYèè CovÀªÀ wvÀÛgÀ YÁ&a*nÀ¼Á&ÀcB gÀzÀcÝ YÀr, ÀvÁnÀcªÁÀzÀc.
4. aªªÁw wozÀc wvÀÛgªÁ&ÀcBªÁÁvÀ wvÀÛgÀ YÁ&a*oÀceè nÀcgÀc&, ÀvÁaÀzÀcÝ. wozª YÁ, ±*Bn* cgÀ&Àc wvÀÛgªÁ&ÀcB nÀcgÀc&, ÀcªÁÀzÀc CªÁÁ&Á.
5. cvÁ! wvÀÛgÀnÀ¼Á&ÀcB eªÀcn* wzÀV, ÀvÁzÀ c.co.Dg@ wvÀÛgÀ YÁ&a*oÀc °Á¼*oÀcªªv* aÀYÀc, CxÀªÁ eÁe ±Á>>oÀc ªv@YÁ-com@ YªeBeozÀ nÀcgÀcvÀcªÁcÁ&ªªÁÀc.

1. ಕರ್ನಾಟಕ ಗತತಿಭವ ಗ್ರಂಥವನ್ನು ರಚಿಸಿದವರು -----
a) & ಲೇರಂಗ b) ಅಲೂರು ಕೆಂಕಟಾಚಾರ್ಯ
c) 8. ಎ. ನರಸಿಂಹರಾಯ d) ಕುಮಾರಪ್ಪ
2. 1955 ರಲ್ಲಿಯೂ ಜೂನ್ 8 ನ ದಿನ ಕನ್ನಡ ಸಾಹಿತ್ಯ ಪರಿಷತ್‌ನ ಸಭೆಯಲ್ಲಿ ಅಧ್ಯಯನ.
a) ಡಿ. ಆರ್. ಕಾರಂತ b) ಚಂಗಲಾಚಾರ್ಯ c) & ಲೇರಂಗ d) & ವಮೋಶ್ವರ
3. ಕನ್ನಡ ಸಾಹಿತ್ಯದ 'ಗಿರಿ' ಎಂಬ ಕಾದಂಬರಿ.
a) 9 ರೇಂಜಿಂಗ್‌ನಿಂದ b) ಸಾಂಪ್ರದಾಯಿಕ
c) & ವಾಚನಿಕ ಕಾರಂತ d) ಎ. ಎ. & ಕೆ.ಎಮ್
4. ಇವು ಸಾಹಿತ್ಯ ಕನ್ನಡದ ಪರಿಚಯ ಎಂದು ಕರೆಯಲಾಗುವ ಕಥಾ ಸಂಕಲನ.
a) ಬೆಂಗಳೂರಿನ b) ಬೆಂಗಳೂರಿನ c) 9 ಕೆ.ಎಮ್‌ನ d) ಜೀವನದ

5. “ಹಾಳಾದಾಕೆ ಉಪೇಂದ್ರನು ಉಪನಿಷತ್‌ಗಳನ್ನು, ತೃಪ್ತಿಯಾದಾಕೆ, ಧರ್ಮದ ಹಳಗಿನಗಳನ್ನು, ಶಯನದ ಧರ್ಮಗಳನ್ನು ಧರ್ಮಗಳನ್ನು ಎಂಬ ವಚನ ಬರೆದವರು.
a) ಬಸವಣ್ಣ b) ಅಕಮಲಾಚಾರ್ಯ c) ಅಲಿದಮಪ, ಭು d) ಧರ್ಮದ ಧರ್ಮಮಯ,
 6. ಕೃಷಿಜನರಾದ ಕೃಷಿರೂಪ ಕನ್ನಡದ ಸ. ಕೃಷಿವಿವರಣೆ ಕೃಷಿ ಪದ, ಗಳು ಕೃಷಿ ಶತಮಾನದ ಕೃಷಿ ಸಂಪಾದನೆ.
a) ಹತ್ತಿರದ b) ಎರಡನೆಯ c) ಆರನೆಯ d) ಒಂಬತ್ತನೆಯ
 7. ಸಹಿ ಸಮಗ್ರ ಮಹಿಮೆ, ಕೃಷಿ -----
a) ಅ8ನೆಯ b) ಎ, ದ, c) ಕೇಕೆ, ಕೆ d) 8ನೆಯ
 8. ಕನ್ನಡ ಬರಹಗಾಗಿ ----- ವರುಷಗಳ ಚರಿತ್ರೆ.
a) 1ನೆಯರ ಬದುಕು b) 1ನೆಯರ ನೂರು
c) ಹತ್ತಿರ 1ನೆಯರ d) 1ನೆಯರ ಎರಡುನೂರು
 9. ಕೃಷಿ ಸಂಪಾದನೆ ಅಧಿಕಾರದ ಬಂದ ವರುಷ -----
a) 1947 b) 1950 c) 1998 d) 1945
 10. “ಕೃಷಿ ಕೃಷಿರೂಪದ ಕೃಷಿಯದ ಕೃಷಿ, ಯಲಂಪದ ಕೃಷಿ ಆಗರದ ಕೃಷಿ 2ನೆಯ ಸ” ಎಂದು ಕೃಷಿ

a) ರನದ b) ಪಂಪ c) ಕುರಿಂಪದ d) ಬಸವಣ್ಣ
 11. ಅಯ್ಯಾ 2ನೆಯ, ನವರ ವಚನದ ಕೃಷಿ ಕೃಷಿ ಅತ, ಂತ ಕೃಷಿ ಮಹತ. ವನದ ಕೃಷಿ ಸಂಪಾದನೆ.
a) ಬಡತನ b) ಕೃಷಿ c) ಬಂಧನ d) ವಜ,
 12. ಗುರು, ಕೃಷಿ ವಚನ ಜಂಗಮ ಈ ಮೂರು ಂತ ಅತ, ಂತ ಕೃಷಿ ಕೃಷಿ
a) ಕೃಷಿ b) 2ನೆಯ c) ನನಿರು d) ಕೃಷಿ 2ನೆಯ
 13. ಅನ, ರನದ ದೂರದ ಕೃಷಿ ಅಂಥವರನದ ಕೃಷಿ ಕೃಷಿ ----- ತಳಗಿನ ಕೃಷಿ.
a) ಸ. ಗುರು b) ಶ, ಮ c) ನರ d) ಬಡತನ

14. ಕುಟುಂಬದೊಳಗೆ ಒಂದು, ಪುನಃ ಪರಿಣಿತ ಮಹಿಳೆ ಎಂದು ಕರೆಯಲ್ಪಡುವವರ ----- ವರ್ಗ
ಶತಾನದೊಳಗೆ ಇದ್ದಾರೆ.
a) 2 b) 4 c) 6 d) 9
15. ಒಡತನ ಸತ್ಯವೆಂದರೆ -----
a) ನನಗಿ b) ಸುಡುಗಿ c) ಸತ್ಯವು d) ವಿನಯವು
16. ಜ್ಞಾನರು ----- ಭಕ್ತರು
a) ಹಿಂದು b) ಇಸ್ರಾಯೆಲ c) ಸರಸ್ವತಿಯು d) ಹಿಂದು
17. ತಲೆಗಿಳಿದು ಕಂಡ, ಮನಸ್ಸಿನಲ್ಲಿ ಎಂಬ ಒಂದು ಒಂದು ಬಿಡುವು.
a) ಪರಂದರವು b) ಬಸವಣ್ಣ c) ಕನಕದಾಸ d) ಅಕಮಲವು
18. ಗುಣ ಕೊಡುವವನು ರಾಯ ಒಂದು ಈ ವಚನವನ್ನು ಬಿಡುವು.
a) ಅಕಮಲವು b) ಅಲಮಪ, ಭು c) ಬಸವಣ್ಣ d) ಜ್ಞಾನವು
19. ಇದು ಇವನು ----- ಒಂದು
a) ಇವನು b) ಜ್ಞಾನ c) ಇವನು d) ಇವನು
20. ----- ಒಂದು ಇದ್ದರೆ ಒಡತನ ಇವನು ಒಂದು ಕಷ್ಟವು
a) ಇವನು b) ತಂದೆಯು c) ಇವನು d) ಇವನು
21. ನಗವು ಸಹಜ ಇವನು ನಗುವುದು -----
a) ಜ್ಞಾನವು b) ಪರಮವು c) ಹಿಂದು d) ಇವನು
22. ಅಂಗವಿಡತನದೊಳಗೆ ಎಂಬುದು ಸಾರ ಒಂದು.
a) ಪರಂದರವು b) ಕನಕದಾಸ c) ದ. ಇ. ಹಿಂದು, d) ಅಕಮಲವು
23. ಇವನು ಎಂಬ ಇವನು ಪರಿಣಿತ ಕು
a) ದ. ಇ. ಹಿಂದು, b) ಬಸವಣ್ಣ c) ಕುಟುಂಬ d) ಅಕಮಲವು

24. ವಾಕ ಪದದ ಅರ್ಥ
- a) ನರಕ b) ಸ. ಗ x c) ಭೂB d) ಮೂA
25. ಮಂಕುಮನಿ ಕೆ ಕೃಷಿಯನ್ನು ಬಿಡವರು.
- a) ಕುತಿಂಪ b) ದ. ರ. ರಿಂಗೆ, c) ಕನಕಾಸ d) 8. 9. ಗ
26. ವಾಕುತಂ ಕವನ ಸಂಗ್ರಹವು ----- ಪರಿಚಿತ.
- a) ಪದಭೂಷಣ b) ಪದಭೂಷಣ c) ವಾನ್ವಿರ d) ಪದ್ಮ, ಿ
27. ಪಾಠ್ಯ ಪದದ ಅರ್ಥ
- a) ಪಾಠ್ಯ b) ಅ' c) ಾ d) ಹಣ
28. ಕನಕಾಸ' ಎಂಬ ವಾನ್ವಿರ ಪರಿಚಿತವನ್ನು ತಿಳಿಸುತ್ತವೆ.
- a) 8. 9. ಗ b) ಕುತಿಂಪ c) ದ. ರ. ರಿಂಗೆ, d) ಕನಕಾಸ
29. ಕುರುಡು ವಾನ್ವಿರವನ್ನು ಕವನಸಂಕಲನವೆಂದು ಆಯ್ಕೆ ಮಾಡಲಾಗಿದೆ.
- a) ವಾಕುತಂ b) ವಾಟಕ c) ಮಂಕುತಮನಿ ಕೆ d) ವಾದನೀ
30. 8. 9. ಗಿಯವರ ವಾಕ್ಯ, ವಾಕ್ಯ
- a) ಮಂಕುಮನಿ b) ಕೂಡಲಸಂಗಮವಿವಿ
c) ರಿನದಮಂವಿವಿ d) ವಾಕ್ಯವಾಕ್ಯ
31. ಎಂಬ ವಾಕ್ಯವು ಕಂಡು ಬಂದಿಲ್ಲ. ರಯ, ಅವರು ಉದಾಹರಿಸಿದ್ದಾರೆ.
- a) ಎಂಬುದು ಕೆ ವಿವಿವಿವಿ b) ಎಂಬುದು ವಿವಿವಿವಿ
c) ಎಂಬುದು ಗಿವಿವಿವಿ d) ಎಂಬುದು ವಿವಿವಿವಿ
32. ವಾ. 9-ರ. ರಯ, ನವರು ----- ಕೆ. ಸ. ಂತ ಇವರಿಂದ 8-ನೇಯನ್ನು ಗಟ್ಟಿಪಡಿಸಿದರು.
- a) 1920 b) 1947 c) 1919 d) 1950
33. ಕ, ಮ, ಸಿ, ತ, ಕಬದಿವಿವಿ ಆಯ್ಕೆ ಮಾಡಿ ಇವು ----- ರವರ ಹುಟ್ಟು, ಗುಣಗಳಿವಿ
- a) ನಾಂವಿವಿ b) ವಿವಿವಿವಿ c) 9-ರ. ರಯ, d) ವಿವಿವಿ

34. ಕೃಷಿ ಸಂಘಟನೆಗಳನ್ನು ಕಟ್ಟಿದವರು ಕಾರು?
- a) 9-ನೇ ಶತಮಾನ, b) ನಾಲ್ಕನೇ ಶತಮಾನ c) 15ನೇ ಶತಮಾನ d) 18ನೇ ಶತಮಾನ
35. ದುರದೃಷ್ಟವಶಾತ್, ಹಳೆಯ ಕೃಷಿ ಸಂಘಟನೆಗಳು ಮತ್ತು ಸೇವೆಗಳು
- a) ಕುಳಿತುಕೊಳ್ಳುವ b) ದ. 18ನೇ ಶತಮಾನ, c) 8. 9. ಶತಮಾನ d) 9-ನೇ ಶತಮಾನ, 18ನೇ ಶತಮಾನ
36. ಕೃಷಿ ಸಂಘಟನೆಗಳು ಕೃಷಿ ಸಂಘಟನೆಗಳನ್ನು ಕಟ್ಟಿದವರು.
- a) 1800 b) 2500 c) 2200 d) 2900
37. ಹಳೆಯ ಕೃಷಿ ಸಂಘಟನೆಗಳು ಕಟ್ಟಿದವರು.
- a) 8. 9. ಗುಂಡಪ್ಪ, b) ಕುಳಿತುಕೊಳ್ಳುವ c) 9-ನೇ ಶತಮಾನ, d) ದ. 18ನೇ ಶತಮಾನ, 18ನೇ ಶತಮಾನ
38. _____ ಬಳಿಯ 2ನೇ ಶತಮಾನದ ಕೃಷಿ ಸಂಘಟನೆ.
- a) ಅಶೋಕ b) ಕೃಷಿ ಸಂಘಟನೆ c) 18ನೇ ಶತಮಾನ d) ಅಶೋಕ
39. ಕೃಷಿ ಸಂಘಟನೆಗಳು ಕಟ್ಟಿದವರು _____ ಕಟ್ಟಿದವರು.
- a) 100 b) 200 c) 400 d) 300
40. ಕೃಷಿ ಸಂಘಟನೆಗಳು ಕಟ್ಟಿದವರು _____ ಕಟ್ಟಿದವರು.
- a) ಭಾರತ, ವಸಿ b) ಕೃಷಿ ಸಂಘಟನೆ c) 18ನೇ ಶತಮಾನ d) ಹಳೆಯ
41. ಕೃಷಿ ಸಂಘಟನೆಗಳು ಕಟ್ಟಿದವರು _____ ಕಟ್ಟಿದವರು.
- a) 18ನೇ ಶತಮಾನ b) 18ನೇ ಶತಮಾನ c) 18ನೇ ಶತಮಾನ d) 18ನೇ ಶತಮಾನ
42. ಕೃಷಿ ಸಂಘಟನೆಗಳು ಕಟ್ಟಿದವರು _____ ಕಟ್ಟಿದವರು.
- a) 18ನೇ ಶತಮಾನ b) 18ನೇ ಶತಮಾನ c) 18ನೇ ಶತಮಾನ d) 18ನೇ ಶತಮಾನ
43. ಕೃಷಿ ಸಂಘಟನೆಗಳು ಕಟ್ಟಿದವರು _____ ಕಟ್ಟಿದವರು.
- a) 18ನೇ ಶತಮಾನ b) 18ನೇ ಶತಮಾನ c) 18ನೇ ಶತಮಾನ d) 18ನೇ ಶತಮಾನ
44. ಕೃಷಿ ಸಂಘಟನೆಗಳು ಕಟ್ಟಿದವರು _____ ಕಟ್ಟಿದವರು.
- a) ಕೃಷಿ ಸಂಘಟನೆ b) 8. 9. ಶತಮಾನ c) 18ನೇ ಶತಮಾನ, d) ಕುಳಿತುಕೊಳ್ಳುವ

45. ಗೋಪಾಲಾಚಾರ್ಯರ 2ನೇ ಪುಸ್ತಕವು _____
 a) ಕನ್ನಡ, b) 2ನೇ ಪುಸ್ತಕ c) ಇಂಗ್ಲಿಷ್ d) 2ನೇ ಪುಸ್ತಕ, 9ನೇ
46. 2ನೇ ಪುಸ್ತಕವು ಹೇಗೆಯಾಗಿ ರಚಿಸಲ್ಪಟ್ಟಿದೆ
 a) ಅಕ್ಷರ b) ಕಥೆ c) ಕುರಿತು d) ಅಕ್ಷರ
47. ಕುರಿತು ಜನಪ್ರಿಯವಾದ ಕಥೆ, _____
 a) ಗೋಪಾಲಾಚಾರ್ಯ b) ತಿರುಗಿಬಂದರು c) ಅಂಥ, ಪ, ಕೆ d) ಮಂಥ, ಪ, ಕೆ
48. 2ನೇ ಪುಸ್ತಕವು ಜನಪ್ರಿಯ, _____
 a) 190 b) 200 c) 180 d) 150
49. ಅಕ್ಷರಗಳನ್ನು 9ನೇ ಪುಸ್ತಕವು ಕುರಿತು 9ನೇ ಪುಸ್ತಕವು ತಿರುಗಿಬಂದರು _____
 a) ಎ. ಕೆ. ಕೆ. b) ಗೋಪಾಲಾಚಾರ್ಯ c) ವೆ. d) 8. 9. ಕೆ
50. '2ನೇ ಪುಸ್ತಕವು ಎಂಬ ಅಕ್ಷರ ಪದವು' ಬರೆದ ಕೆ. _____
 a) ಕನ್ನಡ b) ಅಕ್ಷರ c) ಪದ d) 9. ಗೋಪಾಲಾಚಾರ್ಯ,

* * * * *

Fourth Semester B.E/B.Tech Degree Examination, June/July 2023

±òÅ  - Samskruthika Kannada

Time: 1 hrs.]

[Max. Marks: 50]

ÀsZÀ£*nÀ¼Àc

1. CW! BO YÀ±*BnÀ½nÀS wvÀÛ&1&. YÀ, & YÀ, ±*Bn* wozÀc CoaÁ.
2. c.co.Dg@ wvÀÛgÀ YÀ&a*oÀceè oÀcc.c, @.c&@, Ào6*, °ÁnÀs YÀ±*B YÀ&a*oÀc ±*Á - °oÀc&ÀcB CozÀg* A, B, C CxÀªÁ D oÀc&ÀcB vÀ!, wèzÀov* aÀ&ÁoÀcªÁV nÀcgÀc&, ÀcªÁÀzÀc CªÁ, %7oÀc vªÁªÁ&toÀÁVgÀcvÀÛz*.
3. c.co.Dg@ wvÀÛgÀ YÀ&a*oÀceè enÀ&YÀr¹gÀcªÁ, ÀÛ¼ÁzÀeè ªÁ&7ªÁcÁ&z* °Án*oªcÁ ele CxÀªÁ ªÁ&7ªÁcÁr zÀªÁcÁ>&toÀceè oÀcÁªªÁz*ªÁªÁ vÁ, ÀezÁYèè CovÀªÁ wvÀÛgÀ YÀ&a*nÀ¼Á&ÀcB gÀzÀcÝ YÀr, ÀvÁnÀcªÁÀzÀc.
4. aªªÁw wozÀc wvÀÛgªÁ&ÀcBªÁcÁvÀ wvÀÛgÀ YÀ&a*oÀceè nÀcgÀc&, ÀvÁaÀzÀcÝ. woz* YÀ, ±*Bn* cgÀ&Àc wvÀÛgªÁ&ÀcB nÀcgÀc&, ÀcªÁÀzÀc CªÁ&Á.
5. cvÁ! wvÀÛgÀnÀ¼Á&ÀcB eªÀcn* wzÀV, ÀvÁzÀ c.co.Dg@ wvÀÛgÀ YÀ&a*oÀc °Á¼*oÀcªÁcªv* aÀYÀc, CxÀªÁ eªe ±Á>>oÀc ªÁv@YÁ-com@ YªeBeozÀ nÀcgÀcvÀcªÁcÁ&ªªÁÀc.

1. ಗೋಪಿಪಾಠಾ 2023ರ ಮಗನ ಚಿತ್ರವೇನು?
a) ಅಮೆರಿಕಾ b) ಅಜ್ಞಾನ c) ಪ, 2023 d) ಪ, ಭೂ
2. ಇತಿಹಾಸದ _____ ಇತಿಹಾಸದ 2023ರ
a) 2023 ರೇಖೆ ಇತಿಹಾಸದ b) ಕಂಪ್ಯೂಟರ್ ಇತಿಹಾಸದ
c) 19ನೇ ಇತಿಹಾಸದ d) 2023, 9ನೇ ಇತಿಹಾಸದ
3. ಪದವಿಪಡಿಸಿದ ಚಿತ್ರವು _____
a) ಚಿತ್ರವು b) ಮಮತೆ c) ಅಮೆರಿಕಾ d) ಚಿತ್ರವು
4. ಯಾವುದು ಕೆಳಗಿನವು _____
a) ಕನಕಪಾಠ b) 8.9. ಗ್ರ c) ವಸುಧೇವಿ, d) ಕುರಿತು
5. ಗೋಪಿಪಾಠಾ 2023ರ ವರ್ಷ _____
a) 2023, b) 2023 c) ಇತಿಹಾಸದ d) 2023, 9ನೇ

- Ver - B - 2 of 6

25. ಬಡತನ ಸತತವಾಗಿರುವುದು -----
- a) ನಗರದಲ್ಲಿ b) ಸುಡನ್ c) ಉತ್ತರ ಅಮೆರಿಕ d) ದಕ್ಷಿಣ ಅಮೆರಿಕ
26. ಜನಸಂಖ್ಯೆ ----- ಭಾಗ
- a) ಹೆಚ್ಚುತ್ತಿದೆ b) ಕಡಿಮೆಯಾಗಿದೆ c) ಸರಿಸುಮಾರು d) ಹೆಚ್ಚುತ್ತಿದೆ
27. ತಲೆದೂಗಿದ ಕಂಡ, ಎಲ್ಲಾ ಮನೆಯಲ್ಲಿ ಎಂಬ ಒಂದು ವಿಷಯವನ್ನು ಬಿಡುವರು.
- a) ಪರಿಶುದ್ಧತೆ b) ಬಸವಣ್ಣ c) ಕನಕದಾಸ d) ಅಕ್ಕಮಹಾದೇವಿ
28. ಗುರಿ ಹೊಂದಿದಂತೆಯೇ ರೈತರು ಒಂದು ಈ ವಚನವನ್ನು ಬಿಡುವರು.
- a) ಅಕ್ಕಮಹಾದೇವಿ b) ಅಲಿದಾಸ, ಭು c) ಬಸವಣ್ಣ d) ಜಾಮಿನಿ,
29. ಇದು ಇನ್ನೊಂದು ----- ಒಂದು ವಿಷಯ
- a) ಇನ್ನೊಂದು b) ಜನಸಂಖ್ಯೆ c) ಇನ್ನೊಂದು d) ಜೀವನ
30. ----- ಒಂದು ಇದ್ದು ಬಡತನ ಇನ್ನೊಂದು ಕಡೆಗೆ
- a) ಎಲ್ಲಾ b) ತನ್ನಿ c) ಇನ್ನೊಂದು d) ಜೀವನ
31. ಕೃಷಿಗಾಗಿ ಗತವಿರುವ ಗ್ರಾಮವನ್ನು ರೈತರು -----
- a) ಹಿ, ರಂಗ b) ಅಲ್ಲದೆ ಕೆಲವು
- c) 8. ಎ. ನರಸಿಂಹರಾಯ d) ಕುರಿತು
32. 1955 ರಲ್ಲಿ ಜನಸಂಖ್ಯೆ ನಡೆದ ಕಂಡು, ಪರಿಷತ್ ಸಭೆಯ ಅಧ್ಯಕ್ಷರು.
- a) ಡಿ. ಆರ್. ಕಾರಂತ b) ಚಂಗಲಾಯಣ c) ಹಿ, ರಂಗ d) ವಿಮಲಾಕ್ಷಿ
33. ಕಂಡು ಕೊಡುವ ಗ್ರಾಮವು 'ಗಿರಿ ಇನ್ನೊಂದು' ಎಂದು ಕರೆಯಲಾಗಿದೆ.
- a) ಇನ್ನೊಂದು b) ಸ್ವಯಂಸೇವಕರು
- c) ವಿವಿಧ ಕಾರಂತ d) ಎ. ಎ. ಕೆ. ಕೆ. ಕೆ.
34. ಇದು ಸಹಜವಾಗಿ ಕಂಡು ಪರಿಷತ್ ಎಂದು ಕರೆಯಲಾಗಿದೆ.
- a) ಉತ್ತರ ಭಾರತ b) ಉತ್ತರ ಭಾರತ c) ಇನ್ನೊಂದು d) ಜೀವನ

- [illegible]

44. ವಾಕ ಪದದ ಅರ್ಥ
 a) ನರಕ b) ಸ. ಗ x c) ಭೂB d) ಮಾA
45. ಮಂಕುಹಮಿನ ಕೆ ಕೃಕಯನುರ ಬಿಿದವರು.
 a) ಕುತಿಂಪ b) ದ. ತಿ. ರಿಂರಿ, c) ಕನಕಾಸ d) 8. 9. ಗ
46. ವಾಕುತಂ ಕವನ ಸಂಗ, ಹಿ'-----ಪಶಿ ಬಂರಿ.
 a) ಪದಿಭೂಷಣ b) ಪದಿಭೂಷಣ c) ಂನಾಂರ d) ಪದಿ, ಿ
47. ತಿಂಪುಣ ಪದದ ಅರ್ಥ
 a) ತಿಂಬೂನು b) ಅ&' c) ಿಂ& d) ಹಣ
48. ಕನರಡಿ' ಿದಲ ಂನಾಂರ ಪ, ಶಿಾಯನುರ ತಂದುರಿಂಟಿ,ವರು
 a) 8. 9. ಗ b) ಕುತಿಂಪ c) ದ. ರ. ರಿಂರಿ, d) ಕನಕಾಸ
49. ಕುರುಡು ಿಂಂರಾಣವನುರ ಶಿವ ಕವನಸಂಕಲರಿಂದ ಆಯಿಂಂಳಿಂಳಿ
 a) ವಾಕುತಂಕ b) ವಾಟಕ c) ಮಂಕುತಮಿನ ಕೆ d) ವಾದರಿೇರಿ
50. 8. 9. ಗಿಯವರ ಿಂವ, ವಾಮ
 a) ಮಂಕುಹಮಿ b) ಕೂಡಲಸಂಗಮೇವ
 c) ರಿನರಮ88ಂಜುನ d) ತಿಂಮವಾಧ

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Fourth Semester B.E/B.Tech Degree Examination, June/July 2023

ಉಂಸಾಹಕ ಕನಡ - Samskruthika Kannada

Time: 1 hrs.]

[Max. Marks: 50]

ÀsZÀ£*nÀ¼Àc

1. CW! BO YÀ±*BnÀ½nÀS wvÀÛ&1&. YÀ, & YÀ, ±*Bn* wozÀc CoaÁ.
2. c.co.Dg@ wvÀÛgÀ YÀ&a*oÀceè oÀcc.c, @.c&@, Ào6*, °ÁnÀs YÀ±*B YÀ&a*oÀc ±*Á - °oÀc&ÀcB CozÀg* A, B, C CxÀªÁ D oÀc&ÀcB vÀ!, wèzÀov* aÀ&ÁoÀcªÁV nÀcgÀc&, ÀcªÁÀzÀc CªÁ, %7oÀc vªÁªÁ&toÀÁVgÀcvÀÛz*.
3. c.co.Dg@ wvÀÛgÀ YÀ&a*oÀceè enÀ&YÀr¹gÀcªÁ, ÀÛ¼ÁzÀeè ªÁ&7ªÁcÁ&zª* °Án*oªcÁ ele CxÀªÁ ªÁ&7ªÁcÁrzªªÁªÁ>&toÀceè oÀcÁªªÁzª*ªªÁ vÁ, ÀezÁYèè Covªª wvÀÛgÀ YÀ&a*nª¼Á&ÀcB gÀzÀcÝ YÀr, ÀvÁnÀcªÁÀzÀc.
4. aªªÁw wozÀc wvÀÛgªªÁ&ÀcBªÁcÁvÀ wvÀÛgÀ YÀ&a*oÀceè nÀcgÀc&, ÀvAaÀzÀcÝ. wozª YÀ, ±*Bn* cgÀ&Àc wvÀÛgªªÁ&ÀcB nÀcgÀc&, ÀcªÁÀzÀc CªÁÁ&Á.
5. cvÁ! wvÀÛgÀnª¼Á&ÀcB eªÀcn* wzÀV, ÀvÁzÀ c.co.Dg@ wvÀÛgÀ YÀ&a*oÀc °Á¼*oÀcªªcÁvªª aÀYÀc, CxÀªÁ eªe ±ª>>oÀcªªv@Yª-com@ YªeBeozÀ nÀcgÀcvÀcªªcÁ&ªªªÀc.

1. ನಗವ□ ಸಹಜ ಗಮx ನಸುವ□ದು -----
a) 2ನವ b) ಪರಘಮx c) ಕ್ಷೇವನ d) ಿವರವರ
2. ಅಂ9ಂತನಯದತÃ ಎಂಬುದು ಕೂರ □ವ,ವಾವು.
a) ಪ□ರಂದರ್ರಾಸ b) ಕನಕ್ಾಸ c) ದ. ತಾ. ಁಂೀಂ್ರಿ, d) ಅಕ್ಮ²ಾ್ರೀ9
3. ತಾಷ್ಕಕ9 ಎಂಬ 9ರುದು ಪ□ಿದ ಕ9
a) ದ. ತಾ. ಁಂೀಂ್ರಿ, b) ಬಸವಣÃ c) ಕುಿಂಪ□ d) ಅಕ್ಮ²ಾ್ರೀ9
4. ವಾಕ ಪದದ ಅಥx
a) ನರಕ b) ಸ. ಗx c) ಭೂB d) ಮೂಿ
5. ಮಂಕುಮನ ಕೆ ಕೃಯನುಢ ಬಿದವರು.
a) ಕುಿಂಪ□ b) ದ. ತಾ. ಁಂೀಂ್ರಿ, c) ಕನಕ್ಾಸ d) 8. 9. ಕ್ಷ

6. ವಾಕ್ಯತಂತ್ರ ಕವನ ಸಂಗ್ರಹವು -----ಪ್ರಶ್ನೆ ಬಂಧಿಸಿದೆ.
- a) ಪದಭೂಷಣ b) ಪದಭೂಷಣ c) ವಾಕ್ಯಾಂಶ d) ಪದಭೂಷಣ
7. ಪ್ರಾಕೃತ ಪದದ ಅರ್ಥ
- a) ಪ್ರಾಕೃತ b) ಅರ್ಥ c) ಪ್ರಾಕೃತ d) ಹಣ
8. ಕನ್ನಡದಿಂದ ಹಿಂದಿನ ವಾಕ್ಯಾಂಶ ಪದಗಳನ್ನು ತಿಳಿಸಿರುವವರು
- a) 8. 9. ಗ್ರ b) ಕುಮಾರವಿಠಲ c) ದ. ರ. ಬಿ. ರಾಜ, d) ಕನ್ನಡದ
9. ಕುರುಡು ಪ್ರಾಕೃತವನ್ನು ಕವನಸಂಕಲನವೆಂದು ಅರ್ಥೈಸಿಕೊಳ್ಳುವವರು
- a) ವಾಕ್ಯತಂತ್ರ b) ವಾಕ್ಯ c) ಮಂತ್ರಮನಿಷಿ d) ವಾದಕರೇ
10. 8. 9. ಗ್ರಂಥದ ಪ್ರಾಕೃತ, ವಾಕ್ಯ
- a) ಮಂತ್ರಮನಿಷಿ b) ಕುರುಡುಸಂಗ್ರಹವು c) ಕನ್ನಡಮಂತ್ರಪ್ರಾಕೃತ d) ಪ್ರಾಕೃತವಾದ
11. ಗ್ರಂಥಪ್ರಾಕೃತ ವಾಕ್ಯಾಂಶದ ಮಗನು ಕನ್ನಡವೇನು?
- a) ಪ್ರಾಕೃತ b) ಪ್ರಾಕೃತ c) ಪ್ರಾಕೃತ d) ಪ್ರಾಕೃತ
12. ಇಪ್ರಾಕೃತ _____ ಇಂಗ್ಲಿಷ್‌ನಲ್ಲಿ ವಾಕ್ಯಾಂಶವು
- a) ಪ್ರಾಕೃತ ಕನ್ನಡ ಇಂಗ್ಲಿಷ್ b) ಕನ್ನಡ, ಪ್ರಾಕೃತ ಇಂಗ್ಲಿಷ್ c) 19ನೇ ಇಂಗ್ಲಿಷ್ d) 20ನೇ, 9ನೇ ಇಂಗ್ಲಿಷ್
13. ಪ್ರಾಕೃತದ ಕನ್ನಡದ ಕನ್ನಡದ ಕನ್ನಡದ _____
- a) ಪ್ರಾಕೃತ b) ಮಮತೆ c) ಪ್ರಾಕೃತ d) ಪ್ರಾಕೃತ
14. ಯುಗದ ಕನ್ನಡದ ಬಗ್ಗೆದವರು _____
- a) ಕನ್ನಡದ b) 8. 9. ಗ್ರ c) ವಾಕ್ಯಾಂಶ, d) ಕುಮಾರವಿಠಲ
15. ಗ್ರಂಥಪ್ರಾಕೃತ ವಾಕ್ಯಾಂಶದ ವಾಕ್ಯ _____
- a) ಕನ್ನಡದ, b) ವಾಕ್ಯಾಂಶ c) ಇಂಗ್ಲಿಷ್ d) 20ನೇ, 9ನೇ

16. 2ನಂವರೆ ಹೆಸರಿನ ರವರ ಜನನದ ಸ್ಥಳವು _____
 a) ಅಹಮದಾಬಾದ್ b) ಕಲಬುರಗಿ c) ಕುಣಿಗಲ್ d) ಕೊಡಗು
17. ಕುಣಿಗಲ್ ಜನಪದದ ರವರ ಕವಿತೆ, _____
 a) ನಂದಿಮಾಲಾ b) ತಿರುಳುಗಾಡು c) ಅಂಥ, ಪ, ಜೀವಿ d) ಮಂಥ, ಪ, ಜೀವಿ
18. 2ನಂವರಿಯ ಕುಣಿಗಲ್ ಜನಪದ, _____
 a) 190 b) 200 c) 180 d) 150
19. ಅಹಮದಾಬಾದ್ ಜನಪದದ ಕವಿತೆ, _____
 a) ಎ. ಕೆ. ಕೊಡಗು b) ನಂದಿಮಾಲಾ
 c) ವೆಂಕಟ d) 8. 9. ಕೆ
20. '2ನಂವರೆ ಎಂಬ ಅಹಮದಾಬಾದ್ ಪದವಿ' ಬರೆದ ಕವಿತೆ _____
 a) ಕನಕಲಾಸುರ b) ಅಯ್ಯಾ' ಲಕುಮ್ c) ಪದ್ಮರಂಜನ d) &. 9. ನಂದಿಮಾಲಾ, _____
21. ಕವಿತೆ ಕೊಡಗು ಜನಪದದ ಕವಿತೆ _____
 a) ಎ. ಕೆ. ಕೊಡಗು b) ಎ. ಕೆ. ಕೊಡಗು
 c) ಎ. ಕೆ. ಕೊಡಗು d) ಎ. ಕೆ. ಕೊಡಗು
22. ಕವಿತೆ, _____
 a) 1920 b) 1947 c) 1919 d) 1950
23. ಕ, ಮ, _____
 a) ನಂದಿಮಾಲಾ b) ಜೀವಿ c) 9. ಕೆ. ಕೊಡಗು, d) ವೆಂಕಟ
24. ಕವಿತೆ _____
 a) 9. ಕೆ. ಕೊಡಗು, b) ನಂದಿಮಾಲಾ c) ವೆಂಕಟ d) 8

25. ದುಃಖವನ್ನು ಪರಿಹರಿಸಲು ಸಹಾಯಕವಾಗಿರುವುದು ಏನು?
- a) ಕುತೂಹಲ b) ದ. ಅ. ಉ. ರ. c) 8. 9. ಕ. d) 9. ರ. ಉ. ರ.
26. ಸ್ವಾತಂತ್ರ್ಯದ ಸಂದರ್ಭದಲ್ಲಿ ಸರ್ಕಾರದ ಕಾರ್ಯದ ಕೊಡುಗೆ ಏನು?
- a) 1800 b) 2500 c) 2200 d) 2900
27. ಹೆಚ್ಚಿನ ಸಂಖ್ಯೆಯ ಜನರು ಒಂದು ಸ್ಥಳದಿಂದ ಮತ್ತೆ ಸ್ಥಳಕ್ಕೆ ಬರುವುದು ಏನು?
- a) 8. 9. ಗುಂಡಪ್ಪ b) ಕುತೂಹಲ c) 9. ರ. ಉ. ರ. d) ದ. ಅ. ಉ. ರ.
28. _____ ಬರೆಯುವುದು ಮೊದಲನೆಯ ಕೆಲಸವಾಗಿದೆ.
- a) ಅಕ್ಷರ b) ಸ್ವಾತಂತ್ರ್ಯ c) & ಉ. ರ. d) ಅ, ಕ
29. ಸ್ವಾತಂತ್ರ್ಯದ ಸಂದರ್ಭದಲ್ಲಿ ಬರೆಯುವ ಸಂಖ್ಯೆಯ _____ ಅಕ್ಷರ.
- a) 100 b) 200 c) 400 d) 300
30. ಕೃಷಿ ಮತ್ತು ಉದ್ಯಮ _____ ಉದ್ಯಮ.
- a) ಭೂ, ವಸತಿ b) & ವಸತಿ c) ಉ. ರ. d) ಹೆಚ್ಚಿನ
31. ಆಯ್ಕೆ ಸಂಸದ, ನವರ ವಸತಿ ಸಂಸದ ಸಂಸದ _____ ಅಕ್ಷರ.
- a) ಬಡತನ b) ಉ. ರ. c) ಬಂಧನ d) ವಸತಿ
32. ಗುರು, ಸಂಸದ ಮತ್ತು ಜನರ ಈ ಮೂಲಕ ಅಕ್ಷರ _____ ಉದ್ಯಮ.
- a) ಉ. ರ. b) 2. 3. c) ನ. 4. d) 2. 3. 4.
33. ಅನುಸಾರವಾಗಿ ಅಕ್ಷರವನ್ನು ಚರ್ಚಿಸುವುದು _____ ತಕ್ಷಣವಾಗಿದೆ.
- a) ಸ. ಗ. ರ. b) ಶ. ಮ. ರ. c) ನ. ರ. d) ಬಡತನ
34. ಕುತೂಹಲದಿಂದ ಉ. ರ. ಪರಿಣಾಮವಾಗಿ ಮಹಿಳೆ ಎಂದು ಕೃಷಿ ಸಂಸದ _____ ಉದ್ಯಮ.
- a) 2 b) 4 c) 6 d) 9
35. ಬಡತನ ಸಂಸದ _____ ಉದ್ಯಮ.

a) ನನಗಿ

b) ಸುಡಗಿ

c) ಫಾಯಲು

d) ವೆಗಿಯಲು

36. ગોસરુ ----- બકારુ

a) &ವನ

b) 9ನೇ

c) ಸರಸ. ಹಯ

d) ಹೃಯ

37. ತಲಶಿಗಿದರು ಕಂಡ, ಎಂಗಳ ಮನಠಿ ಎಂಬ ಹೀತಖೆಯನು ಬಾಿದವರು.

a) ಪರಿವರ್ತನಾ

b) ಬಸವಣ್ಣ

c) ಕನಕಜ್ವಲ

d) ಅಕಮ್‌ವಿ಼ಂಞಿಱ

38. ಗುಂಪಿ ಈ ಲಿಪಿಯಲ್ಲಿ ರಿಯ ಒಂದು ಈ ವಚನವನ್ನು ಬರೆದವರು.

a) ಅಕಮ್‌ವಿಃ

b) ಅಲಡಿಮಪ, ಭು

c) ಬಸವಣ್ಣ

d) ಇೌಮಯ,

[illegible]

a) ଅଂବିଧି

b) ಿಠಸನ

c) ಅಂ ಂನ

d) ಿವರ

40.-----ಒಲು2 ಇದAಾಿ ಬಡತನ 9Aಸುವdu ಕಷ್ಯವಲಿ

[illegible]

b) ತಂಜಿಯ

c) ಅಂ ಂನ

d) ಿಂವದ

41. ಕವಾಟಕ ಗತಂಭವ ಗಂಧವನು ರಗದವರು -----

a) & ଲେଖକ

b) ಆಲೂರು ಕೆಂಕಟಾಯ

c) 8. ಎe. ನರಗಿಂಹರಾಯx

d) ಕುಂಞಿಂಪ

42 1955 ರಲ್ಲಿಯೇ ಜೊತೆಗೂಡಿ ನೌಕರರ ಕಲ್ಯಾಣ, ವ್ಯವಹಾರಗಳ ಸುಗಮತೆಯನ್ನು

a) ཁྱིཾ་ ལྟོ་ ཁྱོཾ་

b) ಚಂಗಲಾಯಿತು

c) & ལྷུང་པོ་

d) &ವಮೂೞX ಂೞಿ

43. ಕನ್ನಡ ಕ್ರಾಂತಿಯ ಸಿಯನು 'ಸಿಗಲ' ಂದು ಕಾದವರು.

a) ၄v ဝိဝဿနာ

b) ಸಾಂಖ್ಯವಿಚಾರ

c) &ವಾಂಮು ಂರಂತ

d) ಎಸ್. ಎಸ್. & ಲಿಮಿಟೆಡ್

44. ಥಮಃ ಸಹಜಃ ಕನಃಗರ ಪರಂಪಃ ಎಂದು ಠಾವ ಂಸನ ಕೂಾ ಂೇಳತಃ.

a) ບັນດາປະຊາກອນ

b) ಉಲ್ಲೇಖಿಸಿ

c) 9ಕೆ'ಮಗಳm&ನ

d) ಿವಣನಿಾಯ

Fourth Semester B.E/B.Tech Degree Examination, June/July 2023

ಉಂಸಾಹಕ ಕನಡ - Samskruthika Kannada

Time: 1 hrs.]

[Max. Marks: 50

ÀsZÀ£*nÀ¼Àc

1. CW! BO YÀ±*BnÀ½nÀS wvÀÛ&1&. YÀ, & YÀ, ±*Bn* wozÀc CoaÁ.
2. c.co.Dg@ wvÀÛGà YÀ&a*oAcèe oAcc.c , @.cF@ , Ào6*, °Anàs YÀ±*B YÀ&,a*oAc ±*Ã - °ocÀcFÀcB CozÀg* A, B, C CxÀªÁ D ocÀcFÀcB vÀ!, wèzÀov* aÀ&Áo:ÀcªÁV nÀcgÀc&, ÀcªÁÀzÀc CªÁ, %7oiÀc vªÁªÁÍ&toAÁVgÀcvÀÛz*.
3. c.co.Dg@ wvÀÛGà YÀ&a*oAcèe enÀ&YÀr¹gÀcªÁ, ÀÛ¼ÀzÀeè ªÁ&7ªÁcÁ&zªÁ °An*o:cÃ ele CxÀªÁ ªÁ&7ªÁcÁrzªÁ ªÁcÁ>&toAcèe ocÀcÁªÁÀzªÁªÁ À, yÁ, ÀezÀYè CovÀ°À wvÀÛGà YÀ&,a*nÀ¼ÀÉÀcB gAzÀcÝ YÀr, ÀvÁnÀcªÁÀzÀc.
4. aªÁªAw wozÀc wvÀÛGªÁÉÀcB ªÁcÁvÀ wvÀÛGà YÀ&a*oAcèe nÀcgÀc&, ÀvAaÀzÀcÝ. woz* YÀ, ±*Bn* cgÀ&Àc wvÀÛGªÁÉÀcB nÀcgÀc&, ÀcªÁÀzÀc CªÁÁÉÀ,
5. cvÁ! wvÀÛGÀnÀ¼ÀÉÀcB eªÁcn* wzÀV, ÀvÁzÀ c.co.Dg@ wvÀÛGà YÀ&a*oAc °Á¼*oiÀc ª*cÃv* aAYÀc, CxÀªÁ eÃe ±Á>>oAc ªAv@YÁ-com@ Y*eBeozÀ nÀcgÀcvÀc ªªÁcÁ&ªªÁÀc.

- [illegible]

5. ದುಃಖವಿಗಾಗಿ ಪುನಃ ಪ್ರಯತ್ನಿಸುವುದು ಎಂದು ತಿಳಿದವರು
 a) ಕುತುಂಬಿಲ್ಲ b) ದ. ಶಿವ. ಬೆಂಗಳೂರು, c) 8. 9. ಕೆ. d) 9-ನೇ. ಶಿವ. ರಾಯ,
6. ಕ್ರಾಂತಿಕಾರಿ ರಾವಿ ವಜ್ರವೆಂದ ಕರಕುಶಲ ಕಲೆಗಾಗಿ ಬೆಂಗಳೂರಿನಲ್ಲಿ.
 a) 1800 b) 2500 c) 2200 d) 2900
7. ಹುಡುಗರು ತಿಳಿಸುವುದು ಒಬ್ಬರೂ ಮರ ತಿಳಿಸುವುದು ಎಂದು ತಿಳಿದವರು.
 a) 8. 9. ಗುಂಡಪ್ಪ, b) ಕುತುಂಬಿಲ್ಲ c) 9-ನೇ. ಶಿವ. ರಾಯ, d) ದ. ಶಿವ. ಬೆಂಗಳೂರು,
8. _____ಬೆಂಗಳೂರು 2ನೇನ ಮುದ್ರಣ ಕಲೆಗೆ ಮೂಲಕೆ.
 a) ಅಶ್ವತ್ಥ b) ಕ್ರಾಂತಿ c) & ಲೇಖಕ d) ಆ, ಕೆ
9. ಕ್ರಾಂತಿಕಾರಿ ಬೆಂಗಳೂರು ಬೆಂಗಳೂರು ಬಣ್ಣಗಳನ್ನು 9ನೇನ ಸುಮಾರು _____ ಆಡಗಿ.
 a) 100 b) 200 c) 400 d) 300
10. ಕೃಷಿ ಮತ್ತು ಉಳಿತಾಯ ಕುರಿತು ವಿಷಯ.
 a) ಭಿಕ್ಷು, ವಸತಿ b) & ವಸತಿ c) ಬೆಂಗಳೂರು d) ಹೆಚ್ಚು
11. ಕೃಷಿ ಮತ್ತು ಗತಿಯು ಭವಿಷ್ಯದ ಗುರುತನ್ನು ಕೊಡುವುದು _____
 a) & ಲೇಖಕ b) ಆಲೂರು ಕೆಂಪುಕಟ್ಟೆಯ
 c) 8. ಎ. ನರಸಿಂಹರಾಯ d) ಕುತುಂಬಿಲ್ಲ
12. 1955 ರಲ್ಲಿ ಜಾರಿಯಾದ ನವೀನ ಕನಡಾ ಕೃಷಿ, ಪರಿಷತ್ತಿನ ಸಭೆಯ ಅಧ್ಯಕ್ಷರು.
 a) ಕೆ. ಆರ್. ಕುಮಾರ b) ಚಂಗಲಾಯ ಕೆ. c) & ಲೇಖಕ d) & ವಸತಿ ಕೆ.
13. ಕನಡಾ ಕೃಷಿಯ ಗುರುತನ್ನು 'ಕೆ' ಎಂದು ಕರೆದವರು.
 a) 9ನೇನ ಬೆಂಗಳೂರು b) ಸೆಂಟ್ರಲ್ ಬೆಂಗಳೂರು
 c) & ವಸತಿ ಕುಮಾರ d) ಎ. ಎ. ಕೆ. ಕೆ. ಕೆ.
14. ಕೃಷಿ ಸಂಪನ್ಮೂಲ ಕನಡಾಗೆ ಪರಿಣಿತ ಎಂದು ರಾವಿ ಕೃಷಿ ಕೃಷಿ ತಿಳಿದವರು.
 a) ಬೆಂಗಳೂರು b) ಬೆಂಗಳೂರು c) 9ನೇನ ಕೆ. d) ಕೃಷಿ ಕೃಷಿ

- [illegible]

24. ಯುಗಾಂತ ಕಾಲದಲ್ಲಿ ಬರೆದವರು _____
 a) ಕನಕಾಸ b) 8. 9. ಗ c) ವಸುಧೇಂದ್ರ d) ಕುಮಾರಪ್ಪ
25. ಗೋಪಾಲಾಚಾರ್ಯರ ಹೆಸರು _____
 a) ಶಿವರುದ್ರ b) 2ನೇ ಸಾಹು c) ಇಂದ್ರನಿಧಿ d) 2ನೇ, 9ನೇ
26. 2ನೇ ಹಂತದ ಕೆಲಸವು ಜನರಿಗೆ _____
 a) ಅನುಕೂಲ b) ಕಳಪೆ c) ಕುಣಿ d) ಅನುಕೂಲ
27. ಕುಣಿ ಜನರಿಗಾಗುವ ಕೆಲಸ, _____
 a) ಗೋಪಾಲಾಚಾರ್ಯ b) ತಿರುಕುಳಿ c) ಅಂಕ, ಪ, ಶಿವರುದ್ರ d) ಮಂ, ಪ, ಶಿವರುದ್ರ
28. 2ನೇ ಹಂತದ ಕುಣಿಯ ಜನಸಂಖ್ಯೆ, _____
 a) 190 b) 200 c) 180 d) 150
29. ಅನುಕೂಲ ಕೆಲಸವು 9ನೇ ಹಂತದ ಕುಣಿ ಮತ್ತು 9ನೇ ಹಂತದ ಕೆಲಸವು _____
 a) ಎ. 2ನೇ. ಕುಣಿ b) 2ನೇ ಹಂತದ ಕುಣಿ
 c) 2ನೇ ಹಂತದ ಕುಣಿ d) 8. 9. ಗ
30. '2ನೇ ಹಂತದ ಕುಣಿ' ಎಂಬ ಅನುಕೂಲ ಕೆಲಸವು _____
 a) ಕನಕಾಸ b) ಅನುಕೂಲ c) ಪಂಚರತ್ನ d) 8. 9. ಗೋಪಾಲಾಚಾರ್ಯ,
31. ನಗರ ಸಹಜ ಕುಣಿ ನಗರವು _____
 a) 2ನೇ ಹಂತದ ಕುಣಿ b) ಪಂಚರತ್ನ c) ಕುಣಿ d) 2ನೇ ಹಂತದ ಕುಣಿ
32. ಅನುಕೂಲ ಕೆಲಸವು _____
 a) ಪಂಚರತ್ನ b) ಕನಕಾಸ c) ದ. 2ನೇ. ಹಂತದ ಕುಣಿ d) ಅನುಕೂಲ

33. ತಿಂಪು, ಕೃಷಿ ಎಂಬ 9ರುದು ಪದಗಳ ಕೃತಿ
 a) ದ. ತಿಂಪು, ಕೃಷಿ, b) ಬಸವಣ್ಣ c) ಕುಮಾರವಿಠಲ d) ಅಕ್ಕಮಹಾದೇವಿ
34. ವಾಕ್ಯ ಪದದ ಅರ್ಥ
 a) ನರಕ b) ಸ. ಗಂಧ c) ಭೂಮಿ d) ಮೋಕ್ಷ
35. ಮಂಕುಹುಲಿ ಕೃಷಿಯನ್ನು ಬಿಡುವರು.
 a) ಕುಮಾರವಿಠಲ b) ದ. ತಿಂಪು, ಕೃಷಿ, c) ಕನಕದಾಸ d) 8. 9. ಗು
36. ವಾಕ್ಯತಂತ್ರ ಕವನ ಸಂಗ್ರಹವು ----- ಪದ್ಯ ಬರೆಯಿದೆ.
 a) ಪದ್ಯಭೂಷಣ b) ಪದ್ಯಭೂಷಣ c) ಕನಕದಾಸ d) ಪದ್ಯ, ಕೃ
37. ತಿಂಪು ಪದದ ಅರ್ಥ
 a) ತಿಂಪುನು b) ಅಕ್ಕ c) ಕುಮಾರವಿಠಲ d) ಹಣ
38. ಕನಕದಾಸರು ಬರೆದ ಕವನ ಸಂಗ್ರಹವು ತಂದುಕೊಟ್ಟವರು
 a) 8. 9. ಗು b) ಕುಮಾರವಿಠಲ c) ದ. ತಿಂಪು, ಕೃಷಿ, d) ಕನಕದಾಸ
39. ಕುರುಡು ಕವನ ಸಂಗ್ರಹವನ್ನು ಕವನ ಸಂಕಲನವೆಂದು ಆಯ್ಕೆ ಮಾಡಿದವರು
 a) ವಾಕ್ಯತಂತ್ರ b) ವಾಕ್ಯತಂತ್ರ c) ಮಂಕುಹುಲಿ ಕೃಷಿ d) ವಾದ್ಯವಿಠಲ
40. 8. 9. ಗುಯವರ ಕವನ, ವಾಕ್ಯ
 a) ಮಂಕುಹುಲಿ b) ಕೂಡಲಸಂಗಮವಿಠಲ
 c) ಕನಕದಾಸ d) ತಿಂಪು
41. ಆಯ್ಕೆ 2ನೆಯ, ನವರ ವಚನದ ಕವನ ಸಂಗ್ರಹವು ಅಕ್ಕ, ಕನಕದಾಸ, ಮಹಾದೇವಿ, ವನಕ, 9ನೆಯವರು.
 a) ಬಡತನ b) ಲಿಪಿ c) ಬರಹ d) ವಚನ
42. ಗುರು, ಕೃಷಿ ಜಂಗಮ ಈ ಮೂರನೆಯ ಅಕ್ಕ, ಕನಕದಾಸ, ತಿಂಪು
 a) ಕುಮಾರವಿಠಲ b) ತಿಂಪು c) ನರಕ d) ತಿಂಪು

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21ME42

Fourth Semester B.E. Degree Examination, June/July 2023 Machining Science and Jigs & Fixtures

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define the working principle of lathe. How can you specify a lathe? (06 Marks)
- b. Explain with neat sketch, the working of Radial drilling machine. (08 Marks)
- c. Explain briefly with sketches at any three drilling operation. (06 Marks)

OR

- 2 a. Draw the engine lathe and label the parts and discuss the function of lathe parts. (10 Marks)
- b. Difference between upmilling and down milling process. (05 Marks)
- c. With a neat sketch, explain construction and working of common grinding machine. (05 Marks)

Module-2

- 3 a. Distinguish between orthogonal and oblique cutting with a neat sketch. (06 Marks)
- b. List and explain different types of chips formed in metal cutting process. (08 Marks)
- c. Sketch and explain single point turning tool geometry. (06 Marks)

OR

- 4 a. List and explain different types of cutting tool materials and state their specific applications. (06 Marks)
- b. Draw a merchant's circle diagram, using usual notations and state the assumptions. (08 Marks)
- c. List the various types of cutting fluids used in metal cutting briefly. Explain. (06 Marks)

Module-3

- 5 a. What is machinability? Define machinability index. (04 Marks)
- b. What are the factors affecting on tool life? (06 Marks)
- c. Explain the process of Electroplating. (10 Marks)

OR

- 6 a. Describe the importance of surface finish process. (04 Marks)
- b. With a neat sketch, explain the Honing process. (06 Marks)
- c. Briefly explain powder coating and Galvanizing process. (10 Marks)

Module-4

- 7 a. Explain with neat sketch the process of Abrasive Jet Machining. (10 Marks)
- b. Explain with neat sketch the process of ultrasonic machining process. (10 Marks)

OR

- 8 a. Explain with neat sketch of Electro Discharge machining. (10 Marks)
b. Explain with neat sketch of Laser beam machining. (10 Marks)

Module-5

- 9 a. What are the importances of Jigs and Fixtures in industries? (06 Marks)
b. List the types of Jigs and Fixtures. (06 Marks)
c. What are the materials used to manufacture Jigs and Fixtures? (08 Marks)

OR

- 10 a. What are the factors to be considered to design Jigs and Fixtures? (06 Marks)
b. Briefly explain on Template, Plate, Channel in Jigs. (06 Marks)
c. Briefly explain the importance of fixtures in milling and turning. (08 Marks)

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CBCS SCHEME

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21ME43

Fourth Semester B.E. Degree Examination, June/July 2023 Fluid Mechanics

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain the following terms:
(i) Total pressure (ii) Centre of pressure
(iii) Gauge pressure (iv) Buoyancy (08 Marks)
b. Derive expression for total pressure force and centre of pressure act on a vertical surface immersed in static fluid. (08 Marks)
c. Discuss on fluid pressure measuring devices. (04 Marks)

OR

- 2 a. Explain the Eulerian and Lagrangian method of fluid flow analysis with suitable example. (08 Marks)
b. Derive the 3-dimensional flow continuity equation in cartesian coordinates. (08 Marks)
c. Calculate the velocity of fluid flow at a point (2, 3) if its 2-D flow stream function is given by $\psi = 2xy$. (04 Marks)

Module-2

- 3 a. Derive the Euler's equation of fluid motion and hence deduce Bernoulli's equation. (10 Marks)
b. Derive an expression for discharge through venturimeter. (10 Marks)

OR

- 4 a. Derive expression for discharge through a triangular notch. (10 Marks)
b. A horizontal venturimeter of 20 cm inlet diameter and 10 cm throat diameter is used to measure an oil flow. The discharge of oil through venturimeter is 60 lit/s. Calculate the reading of oil-mercury differential manometer. Take $C_d = 0.98$ and specific gravity = 0.8. (10 Marks)

Module-3

- 5 a. Derive Hagen Poiseuille equation for laminar flow through a circular pipe. (10 Marks)
b. A crude oil flowing through a horizontal circular pipe of 10 cm diameter and 100 cm length. Assume laminar flow and calculate pressure drop if 100 kg oil collected in a tank in 30 seconds. Take viscosity = 0.97 N-S/m² and specific gravity = 0.9. (10 Marks)

OR

- 6 a. Discuss the energy losses that occur in pipe flow. (10 Marks)
b. Derive Darcy-Weisbach equation for determining loss of head due to friction. (10 Marks)

Module-4

- 7 a. Explain the following terms:
(i) Boundary layer thickness (ii) Streamline body (iii) Bluff body
(iv) Lift (v) Drag (10 Marks)

- b. Deduce an expression for pressure drop (Δp) in a pipe flow using Buckingham's π - theorem if fluid has velocity (V), viscosity (μ) and density (ρ). Consider pipe diameter (D) and length (L). (10 Marks)

OR

- 8 a. Explain the following terms:
 (i) Reynold's number (ii) Froude's number (iii) Euler's number
 (iv) Weber's number (v) Mach number (10 Marks)
- b. A flat plate $1.5 \text{ m} \times 1.5 \text{ m}$ moves at 50 km/hr in stationary air of density 1.15 kg/m^3 . The coefficients of drag and lift are 0.15 and 0.75 respectively. Compute:
 (i) Lift force
 (ii) Drag force
 (iii) Resultant force
 (iv) Power required to keep the plate in motion. (10 Marks)

Module-5

- 9 a. Show that velocity of elastic wave propagation in an adiabatic medium is given by $C = \sqrt{\gamma RT}$. (10 Marks)
- b. A projectile travels in air of pressure 100 kPa at 10°C with a speed of 1500 km/hr . Compute the Mach number and Mach angle. Take $\gamma = 1.4$ and $R = 287 \text{ J/kg-K}$. (10 Marks)

OR

- 10 a. Explain the necessity, applications and limitations of CFD. (10 Marks)
- b. A projectile travels with a speed of 1500 km/hr at 20°C temperature and 0.1 MPa air pressure. Calculate the Mach number and Mach angle. Take $\gamma = 1.4$ and $R = 287 \text{ J/kg-K}$. (10 Marks)

Fourth Semester B.E. Degree Examination, June/July 2023

Mechanics and Materials

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define the following terms :
 i) Elasticity ii) Stress iii) Strain iv) Young's modulus v) Poisson's ratio. (05 Marks)
- b. Derive an expression for extension of the uniformly tapered circular bar subjected to an axial load. (05 Marks)
- c. A member ABCD is subjected to point loads P_1 , P_2 , P_3 and P_4 as shown in Fig.Q1(c). Calculate the force P_2 necessary for equilibrium if $P_1 = 45\text{kN}$, $P_3 = 450\text{kN}$ and $P_4 = 130\text{kN}$. Determine stresses in each member and also determine the total elongation of the member assuming the $E = 2.1 \times 10^5 \text{N/mm}^2$.

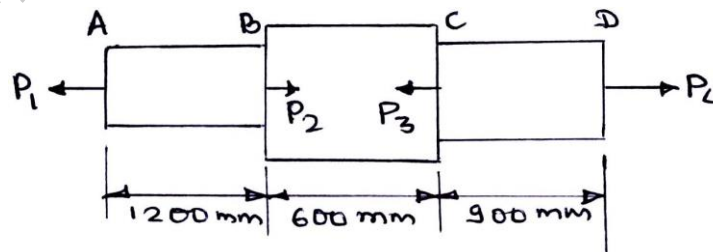


Fig.Q1(c)

(10 Marks)

OR

- 2 a. Derive relationship between modulus of elasticity and modulus of rigidity. (10 Marks)
- b. A compound bar is made up of a central steel plate 50mm wide and 10mm thick to which copper plate 50mm wide and 5mm thick are connected rigidly on each side. The length of the compound bar at room temperature is 1000mm. If the temperature is raised by 100°C , determine the stress in each material and change in length of the compound bar. Assume $E_{\text{st}} = 200\text{GPa}$, $E_{\text{CO}} = 100\text{GPa}$. (10 Marks)

Module-2

- 3 a. Derive an expression for the normal stress and shear stress on a plane inclined at ' θ ' to the vertical axis in a biaxial stress system. (10 Marks)
- b. An element with the stresses acting on it as shown in Fig.Q3(b). Determine :
 i) Principal stresses and its locations
 ii) Maximum shear stresses and its locations.

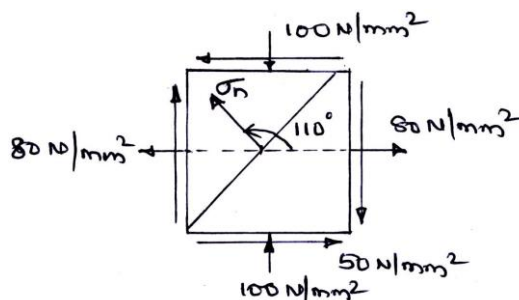


Fig.Q3(b)

(10 Marks)

OR

- 4 The state of stress at a point in a strained material is shown in Fig.Q4. Determine :
- Stresses on a plane whose normal is at an angle of 45° with reference to 80N/mm^2 stress direction
 - Magnitude of principal stresses and their location
 - Maximum and minimum shear stress and their location
 - Draw Mohr's circle and verify the results obtained analytically.

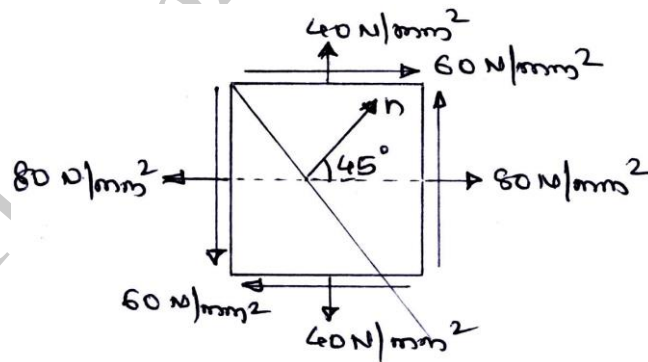


Fig.Q4

(20 Marks)

Module-3

- 5 a. Obtain expressions relating load, shear force and bending moment. (05 Marks)
- b. Draw the shear force and bending moment diagram for the beam shown in Fig.Q5(b).

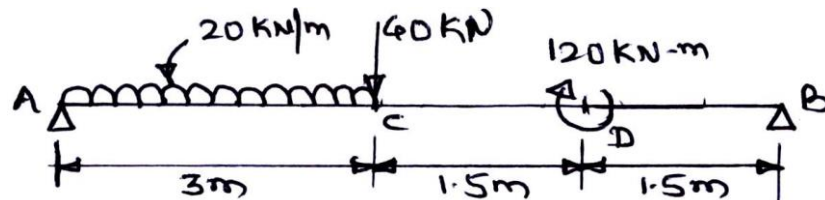


Fig.Q5(b)

(15 Marks)

OR

- 6 a. Derive the equation $\frac{M}{I} = \frac{\sigma_b}{Y} = \frac{E}{R}$ with usual notations. State the assumptions in the derivation. (10 Marks)
- b. A beam having T-section with its flanges of $180\text{mm} \times 10\text{mm}$ and web of $220\text{mm} \times 10\text{mm}$ is subjected to sagging bending moment 15kN-m . Determine the maximum tensile stress and maximum compressive stress, and their location in the section. (10 Marks)

Module-4

- 7 a. Derive differential equation for deflection of beam. (10 Marks)
- b. Determine slope and deflection for a cantilever beam of length L and subjected to UDL $W/\text{unit length}$. (10 Marks)

OR

- 8 a. State assumptions and derive the torsional equation $\frac{T}{J} = \frac{\tau}{R} = \frac{G\theta}{L}$. (10 Marks)
- b. A hollow shaft of diameter ratio 3/8 is required to transmit 588kW at 110rpm, the maximum torque being 120% of the mean. Shear stress is not exceed 63N/mm² and twist in a length of 3m not to exceed 1.4° calculate external diameter of shaft which would satisfy these conditions. Take modulus of rigidity as 84GPa. (10 Marks)

Module-5

- 9 a. Derive an expression for circumferential stress and longitudinal stress for a thin cylinder. (10 Marks)
- b. Derive an expression for strain energy for a member subjected to axial load. (05 Marks)
- c. A steel bar 15mm diameter is pulled axially by a force of 10kN. If the bar is 250mm long, calculate the strain energy stored per unit volume of the bar and total strain energy stored by the bar. Take $E = 2 \times 10^5$ N/mm². (05 Marks)

OR

- 10 a. Obtain the expression for Euler's critical load for a long column with both ends hinged. Also state assumptions made. (10 Marks)
- b. A thick cylinder with internal diameter 80mm and external diameter 120mm is subjected to an external pressure of 40N/mm² when the internal pressure is 120N/mm². Plot the variation of circumferential stress and radial pressure on the thickness of the cylinder. (10 Marks)

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Question Paper Version : A

Fourth Semester B.E./B.Tech. Degree Examination, June/July 2023
Universal Human Values

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

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4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

-
1. Harmony should be maintained in
 - a) Between body and life
 - b) Between self and society
 - c) Between life and environment
 - d) All of these
 2. Basic human aspirations are
 - a) Continuous Happiness
 - b) Prosperity
 - c) Both a and b
 - d) None of these
 3. Purpose of value education is
 - a) Foster universal core values
 - b) Make the syllabus easy
 - c) Develop values in individuals
 - d) All of these
 4. Knowing means having the
 - a) Self exploration
 - b) Self evaluation
 - c) Right understanding
 - d) Having knowledge of all
 5. Happiness means
 - a) To be in a state of harmony
 - b) Fulfillment of desired expectations from others
 - c) Fulfillment of desired feelings from others
 - d) Fulfillment of desired sensations from body
 6. _____ is called foundation value
 - a) Respect
 - b) Affection
 - c) Love
 - d) Trust
 7. The Human goal at the level of nature is
 - a) Prosperity
 - b) Fearlessness
 - c) Co-existence
 - d) Right understanding
 8. The outcome of justice is _____
 - a) Right understanding
 - b) Prosperity
 - c) Trust and Fearlessness
 - d) Coexistence with nature

9. Society means
 - a) Family
 - b) All human beings
 - c) Few individuals
 - d) None of these
10. Feeling for those who have made effort for excellence is _____
 - a) Excellence
 - b) Reverence
 - c) Glory
 - d) None of these
11. An individual people aspiring for the universal human order will be
 - a) More responsible socially and ecologically
 - b) More rich
 - c) More powerful
 - d) More well travel
12. A harmonious world is created by values at 4 levels there are
 - a) Home, Family, Society, Universe
 - b) Individual, Family, Society, Universe
 - c) School, Home, Office, Temple
 - d) None of these
13. "What is the innateness of Tulsi-Plant (or any plant)?
 - a) Existence + Growth
 - b) Cruelty/non-cruelty
 - c) Nurture/Worsen
 - d) Will to live in self
14. As individual people anticipates for the universal human order may be
 - a) More rich
 - b) More Powerful
 - c) More responsible socially and ecologically
 - d) More well-travelled
15. Understanding of human values leads to the practice of
 - a) Responsibilities
 - b) Ethics
 - c) Profession
 - d) Professional ethics
16. "Seeing the self by the self" means
 - a) The consciousness observing the consciousness
 - b) The consciousness observing the material
 - c) The consciousness observing the co-existence
 - d) None of these
17. Education-Right living leads to _____
 - a) Right understanding
 - b) Confusions
 - c) Doubts
 - d) None of these
18. There are _____ dimensions of living
 - a) Six (6)
 - b) Four (4)
 - c) Nine (9)
 - d) Five (5)
19. The second order of nature is _____
 - a) Material order
 - b) Bio order
 - c) Animal order
 - d) Human order
20. The innateness of material order is _____
 - a) Existence
 - b) Cruelty/Non-cruelty
 - c) Growth
 - d) Nurture/Worsen
21. Comprehensive human goal at the level of individual
 - a) Prosperity
 - b) Fearlessness
 - c) Co-existence
 - d) Right understanding
22. The fourth order of nature is
 - a) Material
 - b) Bio order
 - c) Animal order
 - d) Human order

23. _____ is the basic unit of human society.
 a) Group b) Individual c) Nature d) Society
24. Which of the following is a positive value?
 a) Self respect b) Anger
 c) Fear d) Narrow mindedness
25. What is the nature of self?
 a) Conscious b) Physio-Chemical
 c) Biochemical d) Semi-Conscious
26. Self exploration is seeing beyond
 a) Universe b) House c) Box d) Your senses
27. Human to human interaction is called as
 a) Work b) Behaviour c) System d) None of these
28. "All are our own, all are interconnected, interdependent" means.
 a) Oneness b) Worship c) Ease d) None of these
29. Prosperity means
 a) Feeling of having more than required physical facility
 b) Feeling of having less physical facility
 c) Feeling of having more physical facility
 d) Deprived of physical facility
30. Realization is the activity of
 a) Self b) Body
 c) Both self and Body d) None of these
31. Who is responsible for happiness and unhappiness?
 a) Self b) Body
 c) Outside situation d) Society
32. Value education helps to
 a) Removes our contradictions
 b) Remove our confusions
 c) Bring harmony at all levels of human living
 d) All of these
33. Undivided society is ensured by
 a) Right understanding b) Physical facilities
 c) Relationship with right understanding d) None of these
34. The third order of nature is _____
 a) Material order b) Bio order
 c) Animal order d) Human order
35. Which is the complete value?
 a) Love b) Respect c) Trust d) Care
36. Body is a _____ unit while the self is a _____ unit.
 a) Material, Consciousness b) Consciousness, Material
 c) Material, Material d) Consciousness, Consciousness

37. Society is an extension of _____
 a) Human Being b) Family c) Nature d) Existence
38. In value education Sanyam means _____
 a) Self-exploration b) Self-evolution
 c) Self-regulation d) None of these
39. To fulfill human aspirations _____ are necessary
 a) Both values and skills b) Values
 c) Skills d) None of these
40. The innateness of Bio order is _____
 a) Existence b) Cruelty / Non-cruelty
 c) Growth d) Nurture / Worsen
41. The feelings for those who have made effort for my excellence is _____
 a) Excellence b) Reverence c) Gratitude d) Glory
42. There are _____ comprehensive human goals
 a) Eight (8) b) Six (6) c) Four (4) d) Nine (9)
43. The first order of nature is _____
 a) Material order b) Bio order
 c) Animal order d) Human order
44. Developing ethical competence in the profession is the only effective way to ensure
 a) Responsibilities b) Ethics
 c) Profession d) Professional ethics
45. Ensuring justice in relationship, on the basis of values leads to _____ in society.
 a) Fearlessness b) Trust
 c) Fearlessness and Trust d) None of these
46. Human goal → match correctly
 A Right understanding 1 Individual
 B Prosperity 2 Family
 C Fearlessness (Trust) 3 Society
 D Co-existence 4 Nature/Existence
 a) 1 – D, 2 – A, 3 – B, 4 – C b) 1 – B, 2 – C, 3 – D, 4 – A
 c) 1 – A, 2 – B, 3 – C, D – 4 d) 1 – C, 2 – D, 3 – A, 4 – B
47. The human goal at the level of family is
 a) Prosperity b) Fearlessness
 c) Co-existence d) Right understanding
48. Sah-astitua means _____
 a) Co-existence b) Co-operation
 c) Co-option d) Corporate identity
49. Acceptance of excellence in others is called
 a) Reverence b) Guidance
 c) Glory d) Gratitude
50. Each human being is co-existence of
 a) Spirit and Sanyam b) Health and prosperity
 c) Self and Body d) Mind and Soul

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Question Paper Version : B

Fourth Semester B.E./B.Tech. Degree Examination, June/July 2023 Universal Human Values

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

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-
1. The feelings for those who have made effort for my excellence is _____
 a) Excellence b) Reverence c) Gratitude d) Glory
 2. There are _____ comprehensive human goals
 a) Eight (8) b) Six (6) c) Four (4) d) Nine (9)
 3. The first order of nature is _____
 a) Material order b) Bio order
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 4. Developing ethical competence in the profession is the only effective way to ensure
 a) Responsibilities b) Ethics
 c) Profession d) Professional ethics
 5. Ensuring justice in relationship, on the basis of values leads to _____ in society.
 a) Fearlessness b) Trust
 c) Fearlessness and Trust d) None of these
 6. Human goal → match correctly

A Right understanding	1 Individual
B Prosperity	2 Family
C Fearlessness (Trust)	3 Society
D Co-existence	4 Nature/Existence

 a) 1 – D, 2 – A, 3 – B, 4 – C b) 1 – B, 2 – C, 3 – D, 4 – A
 c) 1 – A, 2 – B, 3 – C, D – 4 d) 1 – C, 2 – D, 3 – A, 4 – B
 7. The human goal at the level of family is
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 8. Sah-astitua means _____
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11. Who is responsible for happiness and unhappiness?
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 - c) Outside situation
 - d) Society
12. Value education helps to
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 - b) Remove our confusions
 - c) Bring harmony at all levels of human living
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14. The third order of nature is _____
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16. Body is a _____ unit while the self is a _____ unit.
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 - c) Material, Material
 - d) Consciousness, Consciousness
17. Society is an extension of _____
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 - c) Nature
 - d) Existence
18. In value education Sanyam means _____
 - a) Self-exploration
 - b) Self-evolution
 - c) Self-regulation
 - d) None of these
19. To fulfill human aspirations _____ are necessary
 - a) Both values and skills
 - b) Values
 - c) Skills
 - d) None of these
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 a) Oneness b) Worship c) Ease d) None of these
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 a) Feeling of having more than required physical facility
 b) Feeling of having less physical facility
 c) Feeling of having more physical facility
 d) Deprived of physical facility
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 a) Self b) Body
 c) Both self and Body d) None of these

* * * * *

Fourth Semester B.E./B.Tech. Degree Examination, June/July 2023
Universal Human Values

Time: 1 hr.]

[Max. Marks: 50

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 - Self
 - Body
 - Outside situation
 - Society
- Value education helps to
 - Removes our contradictions
 - Remove our confusions
 - Bring harmony at all levels of human living
 - All of these
- Undivided society is ensured by
 - Right understanding
 - Physical facilities
 - Relationship with right understanding
 - None of these
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 - Bio order
 - Animal order
 - Human order
- Which is the complete value?
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 - Respect
 - Trust
 - Care
- Body is a _____ unit while the self is a _____ unit.
 - Material, Consciousness
 - Consciousness, Material
 - Material, Material
 - Consciousness, Consciousness
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 - Human Being
 - Family
 - Nature
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- In value education Sanyam means _____
 - Self-exploration
 - Self-evolution
 - Self-regulation
 - None of these

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 - c) Skills
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 - a) Between body and life
 - b) Between self and society
 - c) Between life and environment
 - d) All of these
12. Basic human aspirations are
 - a) Continuous Happiness
 - b) Prosperity
 - c) Both a and b
 - d) None of these
13. Purpose of value education is
 - a) Foster universal core values
 - b) Make the syllabus easy
 - c) Develop values in individuals
 - d) All of these
14. Knowing means having the
 - a) Self exploration
 - b) Self evaluation
 - c) Right understanding
 - d) Having knowledge of all
15. Happiness means
 - a) To be in a state of harmony
 - b) Fulfillment of desired expectations from others
 - c) Fulfillment of desired feelings from others
 - d) Fulfillment of desired sensations from body
16. _____ is called foundation value
 - a) Respect
 - b) Affection
 - c) Love
 - d) Trust
17. The Human goal at the level of nature is
 - a) Prosperity
 - b) Fearlessness
 - c) Co-existence
 - d) Right understanding
18. The outcome of justice is _____
 - a) Right understanding
 - b) Prosperity
 - c) Trust and Fearlessness
 - d) Coexistence with nature
19. Society means
 - a) Family
 - b) All human beings
 - c) Few individuals
 - d) None of these
20. Feeling for those who have made effort for excellence is _____
 - a) Excellence
 - b) Reverence
 - c) Glory
 - d) None of these
21. The feelings for those who have made effort for my excellence is _____
 - a) Excellence
 - b) Reverence
 - c) Gratitude
 - d) Glory
22. There are _____ comprehensive human goals
 - a) Eight (8)
 - b) Six (6)
 - c) Four (4)
 - d) Nine (9)
23. The first order of nature is _____
 - a) Material order
 - b) Bio order
 - c) Animal order
 - d) Human order

24. Developing ethical competence in the profession is the only effective way to ensure
 a) Responsibilities b) Ethics
 c) Profession d) Professional ethics
25. Ensuring justice in relationship, on the basis of values leads to _____ in society.
 a) Fearlessness b) Trust
 c) Fearlessness and Trust d) None of these
26. Human goal → match correctly
 A Right understanding 1 Individual
 B Prosperity 2 Family
 C Fearlessness (Trust) 3 Society
 D Co-existence 4 Nature/Existence
 a) 1 – D, 2 – A, 3 – B, 4 – C b) 1 – B, 2 – C, 3 – D, 4 – A
 c) 1 – A, 2 – B, 3 – C, D – 4 d) 1 – C, 2 – D, 3 – A, 4 – B
27. The human goal at the level of family is
 a) Prosperity b) Fearlessness
 c) Co-existence d) Right understanding
28. Sah-astitua means _____
 a) Co-existence b) Co-operation
 c) Co-option d) Corporate identity
29. Acceptance of excellence in others is called
 a) Reverence b) Guidance
 c) Glory d) Gratitude
30. Each human being is co-existence of
 a) Spirit and Sanyam b) Health and prosperity
 c) Self and Body d) Mind and Soul
31. Comprehensive human goal at the level of individual
 a) Prosperity b) Fearlessness
 c) Co-existence d) Right understanding
32. The fourth order of nature is
 a) Material b) Bio order c) Animal order d) Human order
33. _____ is the basic unit of human society.
 a) Group b) Individual c) Nature d) Society
34. Which of the following is a positive value?
 a) Self respect b) Anger
 c) Fear d) Narrow mindedness
35. What is the nature of self?
 a) Conscious b) Physio-Chemical
 c) Biochemical d) Semi-Conscious
36. Self exploration is seeing beyond
 a) Universe b) House c) Box d) Your senses
37. Human to human interaction is called as
 a) Work b) Behaviour c) System d) None of these

38. "All are our own, all are interconnected, interdependent" means.
 a) Oneness b) Worship c) Ease d) None of these
39. Prosperity means
 a) Feeling of having more than required physical facility
 b) Feeling of having less physical facility
 c) Feeling of having more physical facility
 d) Deprived of physical facility
40. Realization is the activity of
 a) Self b) Body
 c) Both self and Body d) None of these
41. An individual people aspiring for the universal human order will be
 a) More responsible socially and ecologically b) More rich
 c) More powerful d) More well travel
42. A harmonious world is created by values at 4 levels there are
 a) Home, Family, Society, Universe b) Individual, Family, Society, Universe
 c) School, Home, Office, Temple d) None of these
43. "What is the innateness of Tulsi-Plant (or any plant)?
 a) Existence + Growth b) Cruelty/non-cruelty
 c) Nurture/Worsen d) Will to live in self
44. As individual people anticipates for the universal human order may be
 a) More rich b) More Powerful
 c) More responsible socially and ecologically d) More well-travelled
45. Understanding of human values leads to the practice of
 a) Responsibilities b) Ethics
 c) Profession d) Professional ethics
46. "Seeing the self by the self" means
 a) The consciousness observing the consciousness
 b) The consciousness observing the material
 c) The consciousness observing the co-existence
 d) None of these
47. Education-Right living leads to _____
 a) Right understanding b) Confusions
 c) Doubts d) None of these
48. There are _____ dimensions of living
 a) Six (6) b) Four (4) c) Nine (9) d) Five (5)
49. The second order of nature is _____
 a) Material order b) Bio order
 c) Animal order d) Human order
50. The innateness of material order is _____
 a) Existence b) Cruelty/Non-cruelty
 c) Growth d) Nurture/Worsen

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Question Paper Version : D

Fourth Semester B.E./B.Tech. Degree Examination, June/July 2023 Universal Human Values

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

-
1. Comprehensive human goal at the level of individual

a) Prosperity	b) Fearlessness
c) Co-existence	d) Right understanding
 2. The fourth order of nature is

a) Material	b) Bio order	c) Animal order	d) Human order
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 3. _____ is the basic unit of human society.

a) Group	b) Individual	c) Nature	d) Society
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 4. Which of the following is a positive value?

a) Self respect	b) Anger
c) Fear	d) Narrow mindedness
 5. What is the nature of self?

a) Conscious	b) Physio-Chemical
c) Biochemical	d) Semi-Conscious
 6. Self exploration is seeing beyond

a) Universe	b) House	c) Box	d) Your senses
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 7. Human to human interaction is called as

a) Work	b) Behaviour	c) System	d) None of these
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 8. "All are our own, all are interconnected, interdependent" means.

a) Oneness	b) Worship	c) Ease	d) None of these
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 9. Prosperity means

a) Feeling of having more than required physical facility	b) Feeling of having less physical facility
c) Feeling of having more physical facility	d) Deprived of physical facility

10. Realization is the activity of
 a) Self
 b) Body
 c) Both self and Body
 d) None of these
11. The feelings for those who have made effort for my excellence is _____
 a) Excellence
 b) Reverence
 c) Gratitude
 d) Glory
12. There are _____ comprehensive human goals
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 D Co-existence 4 Nature/Existence
 a) 1 – D, 2 – A, 3 – B, 4 – C
 b) 1 – B, 2 – C, 3 – D, 4 – A
 c) 1 – A, 2 – B, 3 – C, D – 4
 d) 1 – C, 2 – D, 3 – A, 4 – B
17. The human goal at the level of family is
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18. Sah-astitua means _____
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 a) Reverence
 b) Guidance
 c) Glory
 d) Gratitude
20. Each human being is co-existence of
 a) Spirit and Sanyam
 b) Health and prosperity
 c) Self and Body
 d) Mind and Soul
21. Who is responsible for happiness and unhappiness?
 a) Self
 b) Body
 c) Outside situation
 d) Society
22. Value education helps to
 a) Removes our contradictions
 b) Remove our confusions
 c) Bring harmony at all levels of human living
 d) All of these

Version – D – 4 of 4

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18ME61

Sixth Semester B.E. Degree Examination, June/July 2023

Finite Element Methods

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define FEM. List the advantages and disadvantages of FEM. (10 Marks)
- b. List and explain steps in FEM. (10 Marks)

OR

- 2 a. Explain simplex, complex and multiplex elements. (06 Marks)
- b. A cantilever beam of span 'L' is subjected to a point load at its free end as shown in Fig.Q2(b). Derive an equation for the deflection at free end by using RR method. Assume polynomial displacement function.

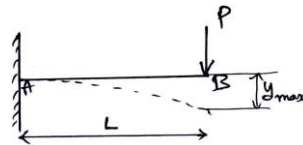


Fig.Q2(b)

(14 Marks)

Module-2

- 3 a. Derive the element stiffness matrix of 1D bar element. (08 Marks)
- b. Using penalty method of handling boundary condition, determine the nodal displacement, stress in each element and support reaction in the bar shown due to applied load in Fig.Q3(b). $P = 100 \text{ kN}$.

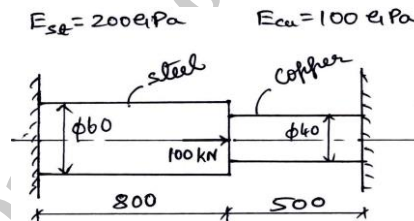


Fig.Q3(b)

(12 Marks)

OR

- 4 a. List the assumptions made in Truss. (04 Marks)
- b. A 4 bar truss element is shown in Fig.Q4(b). Determine the following:
 - i) Nodal displacement
 - ii) Stress in each element
 - iii) Reaction at supports.
 Area of each truss element = 100 mm^2 ; $E = 2 \times 10^5 \text{ N/mm}^2$.

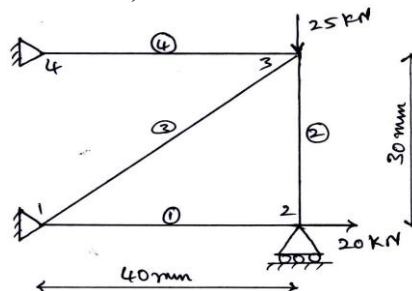


Fig.Q4(b)

(16 Marks)

Module-3

- 5 a. Derive the Hermite shape function of a Beam element. (08 Marks)
 b. For the beam shown in Fig.Q5(b), determine the displacement at node 2 and internal loads. Take $E = 210 \text{ GPa}$, $b = 0.2 \text{ m}$; $h = 0.4 \text{ m}$.

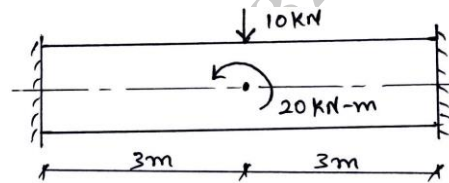


Fig.Q5(b)

(12 Marks)

OR

- 6 a. Derive the stiffness matrix for the torsion of shafts. (08 Marks)
 b. A solid stepped bar of circular cross section shown in Fig.Q6(b) is subjected to a torque of 1 kN-m at its free end and a torque of 3 kN-m at its change in c/s. Determine the angle of twist and shear stresses in the bar. Take $E = 2 \times 10^5 \text{ N/mm}^2$ and $G = 7 \times 10^4 \text{ N/mm}^2$.

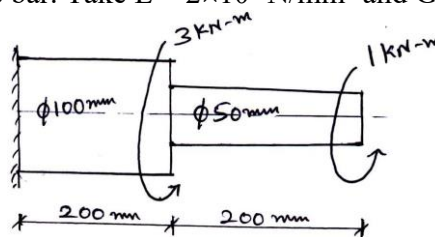


Fig.Q6(b)

(12 Marks)

Module-4

- 7 a. Derive the governing differential equation for 1-D heat conduction. (06 Marks)
 b. Determine the temperature distribution in the composite wall using 1D heat elements, use penalty approach of handling BC's. Refer Fig.Q7(b).

Given :

$$\begin{aligned} k_1 &= 20 \text{ W/m } ^\circ\text{C} \\ k_2 &= 30 \text{ W/m } ^\circ\text{C} \\ k_3 &= 55 \text{ W/m } ^\circ\text{C} \\ h &= 30 \text{ W/m}^2 \text{ } ^\circ\text{C} \\ T_\infty &= 900^\circ\text{C} \\ A &= \text{Unit area} \end{aligned}$$

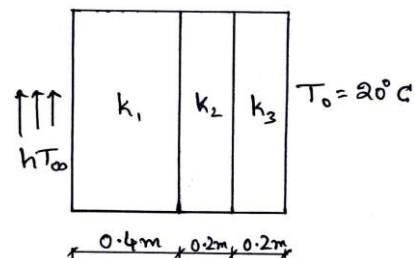


Fig.Q7(b)

(14 Marks)

OR

- 8 a. Derive the element stiffness matrix of 1-D fluid flow element. (06 Marks)
 b. For the smooth pipe of variable cross-section shown in Fig.Q8(b), determine the potential at the junction, the velocities in each section of pipe and the volumetric flow rate. The potential at left end is $P_1 = 10 \text{ m}^2/\text{s}$ and at right end is $P_4 = 1 \text{ m}^2/\text{s}$. For the fluid flow through a smooth pipe $k_x = 1$.

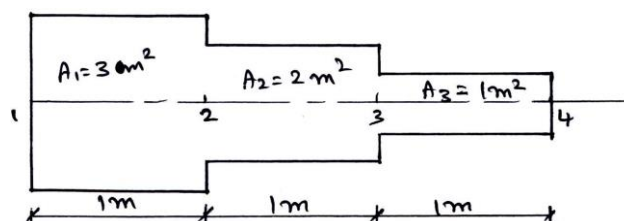


Fig.Q8(b)

(14 Marks)

Module-5

- 9 a. Derive the element stiffness matrix of a triangular axisymmetric element using potential energy approach. (06 Marks)
- b. For the element of an axisymmetric body rotating with constant angular velocity $\omega = 1000 \text{ rev/min}$ as shown in Fig.Q9(b). Determine the body force vector. Include the weight of the material, where specific density is 7850 kg/m^3 .

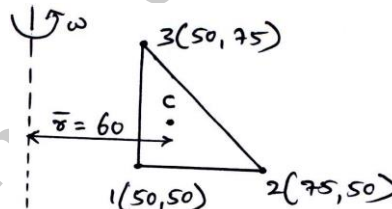


Fig.Q9(b)

(14 Marks)

OR

- 10 a. Derive an expression of element mass matrices of
 (i) 1-D bar element
 (ii) Truss element (10 Marks)
- b. Evaluate eigen value and eigen vector of longitudinal vibration of the constrained uniform circular bar shown in Fig.Q10(b). Take minimum two elements. Take $E = 210 \text{ GPa}$ and $\rho = 7860 \text{ kg/m}^3$.

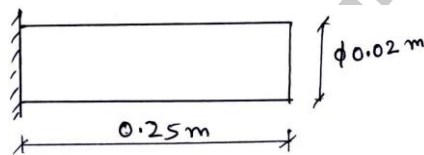


Fig.Q10(b)

(10 Marks)

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Sixth Semester B.E. Degree Examination, June/July 2023 Design of Machine Elements – II

Time: 3 hrs.

Max. Marks: 100

- Note:** 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. Using design data hand book is permitted.
3. Assume missing data suitably.

Module-1

- 1 a. Discuss about the following terms :
 - (i) Active coils
 - (ii) Deflection
 - (iii) Solid length
 - (iv) Free length
 - (v) Resilience (05 Marks)
- b. Derive an expression for energy stored in a spring. (05 Marks)
- c. Design a helical compression spring to carry a load of 500 N with a deflection of 20 mm. The allowable shear stress in the spring material is 350 MN/m² and the modulus of rigidity is 82.7×10^3 MN/m². The spring index is 6. (10 Marks)

OR

- 2 a. A leather belt 125 mm wide and 6 mm thickness transmits power from a pulley 750 mm diameter which runs at 500 rpm. The angle of lap is 150° and the coefficients of friction between the belt and the pulley is 0.3. If the belt density is 1000 kg/m³ and the stress in the belt is not to exceed 2.75 N/mm², find the power that can be transmitted by the belt. Also find the initial tension in the belt. (10 Marks)
- b. An oil well has to be drilled to a depth of 900 mm using 100 drill pipe. Assume 200 N for every 15 m length of pipe. The rope sheaves are of 80 mm diameter and acceleration is 2.5 m/s². Determine the size of 6×37 wire rope for lifting the string of pipes using a FOS as 3 and ultimate stress as 1800 MPa. (10 Marks)

Module-2

- 3 Design a pair of spur gear to transmit 27 kW for an oil pump with the gear ratio of 3 : 1, the rpm of the pinion is 1200, the centre distance is 400 mm, and the gears are to be forged steel untreated with $14\frac{1}{2}^\circ$ FDI. Check the design for dynamic and wear condition. (20 Marks)

OR

- 4 A pair of helical gears are used to transmit 15 kW. The teeth are 20° full depth in normal plane and have a helix angle of 30°. The pinion has 24 teeth and operates at 1000 rpm. The velocity ratio is 5 to 1. The pinion is made of cast steel [$\sigma_d = 50$ MPa] and the gear is of bronze [$\sigma_d = 40$ MPa]. The pinion material is hardened to 200 BHN. Design the gear pair. (20 Marks)

Module-3

- 5 A pair of straight tooth bevel gear at right angle is to transmit 5 kW at 1200 rpm of the pinion. The diameter of the pinion is 80 mm and the velocity ratio is 3.5 to 1. The tooth form is $14\frac{1}{2}^\circ$ composite type. Both pinion and gear are made of CI $[\sigma_d = 55 \text{ N/mm}^2]$. Determine the face width and the required module from the stand point of strength using Lewis equation and check for design from the stand point of dynamic load and wear load. (20 Marks)

OR

- 6 Design a worm gear to transmit 2 kW at 1000 rpm, speed ratio is 20 and centre distance is 200 mm. (20 Marks)

Module-4

- 7 a. A cone clutch with a face angle of 14° has to transmit 286 N-m of torque at a speed of 600 rev/min. The larger diameter of the clutch is 250 mm, face width is 60 mm and co-efficient of friction is 0.18. Determine (i) Axial force to transmit the torque (ii) Average normal pressure (iii) Maximum normal pressure. Assume uniform wear condition. (10 Marks)
- b. A single plate friction clutch of both sides effective has 0.3 m outer diameter and 0.16 m inner diameter. The coefficient of friction is 0.2 and it runs at 1000 rpm. Find the power transmitted for uniform wear and uniform pressure distribution cases if the allowable maximum pressure is 0.08 MPa. (10 Marks)

OR

- 8 a. Fig. Q8 (a) shows a CI brake shoe. The coefficient of friction is 0.30. The breaking torsional moment is to be 346 N. Determine
- The force P, for anti-clock wise rotation.
 - The force P, for clockwise direction.
 - Where must the pivot be placed to make the brake self energizing with the counter clockwise direction.

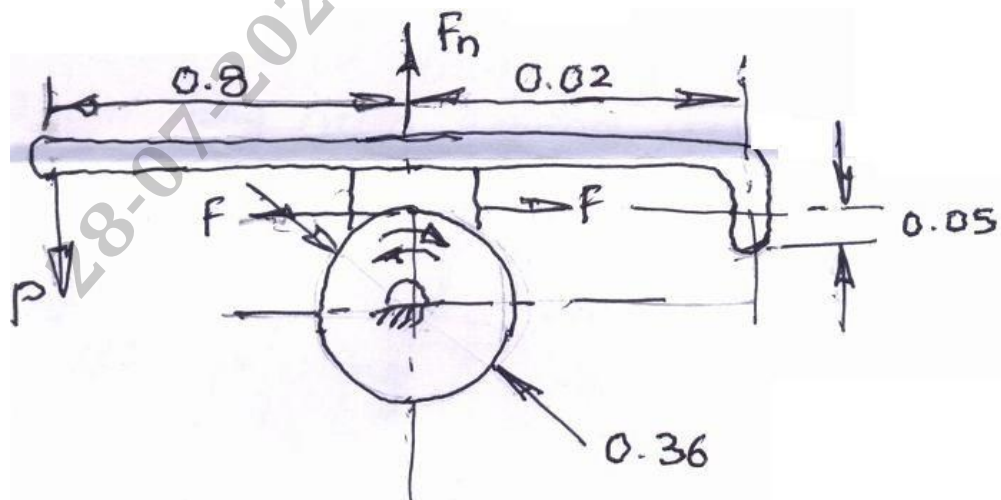


Fig. Q8 (a)

(10 Marks)

- b. In a simple band brake, the length of the lever is 440 mm, the tight end of the band is attached to the fulcrum of the lever and the slack end to a pin 50 mm from the fulcrum. The diameter of the brake drum is 1 m and arc of contact is 300° , the co-efficient of friction between the band and the drum is 0.35. the brake drum is attached to a hoisting drum of diameters 0.65 m that sustains a load of 20 kN (Fig. Q8(b)),
- Force required at the end of lever to support the load.
 - Width of steel band if the tensile stress is limited to 50 N/mm^2 .

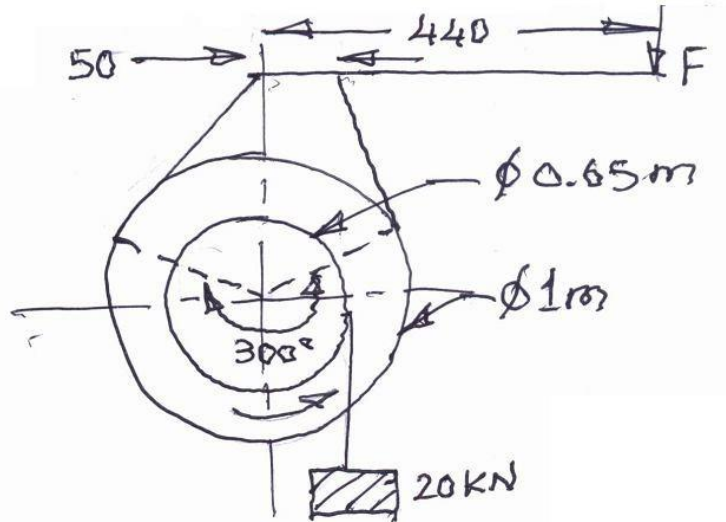


Fig. Q8 (b)

(10 Marks)

Module-5

- 9 a. Derive Petroff's equation for lightly loaded bearing. (12 Marks)
- b. For a full journal bearing has the following specification : Shaft diameter 45 mm, bearing length 66 mm, Clearance ratio 0.0015, Speed 2800 rpm, Load 800 N and absolute viscosity $8.27 \times 10^{-3} \text{ Pa}\cdot\text{s}$. Determine (a) frictional torque (b) Co-efficient of friction (c) Power loss. (08 Marks)

OR

- 10 a. A full journal bearing of 50 mm diameter and 100 mm long has a bearing pressure of 1.4 N/mm^2 . The speed of journal is 900 rpm and the ratio of journal diameter to the diametrical clearance is 1000. The bearing is lubricated with oil whose absolute viscosity at the operating temperature of 75°C may be taken as 0.011 kg/m . The room temperature is 35°C . Determine :
- The amount of artificial cooling required.
 - The mass of lubricating oil required, if the difference between outlet and inlet temperature of the oil is 10°C .
- Take specific heat of $1850 \text{ J/kg}^\circ\text{K}$. (10 Marks)
- b. A bearing for an axial flow compressor is to carry a radial load of 4905 N and thrust load of 2452 N. The service imposes light shock and the bearing is used for 40 hours/week for 5 years. The speed of the shaft is 300 rpm and diameter of the shaft is 60 mm. Select a suitable bearing. (10 Marks)

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18ME63

Sixth Semester B.E. Degree Examination, June/July 2023

Heat Transfer

Time: 3 hrs.

Max. Marks: 100

- Note:** 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. Use of Thermodynamic and Heat Transfer data hand book is permitted.
3. Any missing data can be suitably assumed.

Module-1

- 1 a. With usual notations, starting from 3-dimensional conduction equation, derive one dimensional equation in rectangular coordinates. (10 Marks)
b. Explain the different boundary conditions as applicable to heat transfer analysis. (10 Marks)

OR

- 2 a. What is critical thickness of insulation? Derive an expression for critical radius of insulation in terms of thermal conductivity and HTC 'h'. (10 Marks)
b. A furnace wall is made up of inside silica brick ($K = 1.856 \text{ W/m-K}$) and outside magnesite brick ($K = 5.568 \text{ W/m-K}$) each 10 mm thick. If inner and outer surface temperature of wall are 820°C and 120°C . Find the heat flow rate through the plane W/m^2 . Take the contact resistance of $1.722 \times 10^{-3} \text{ m}^2\text{-K/W}$. Also find the interface temperature. (10 Marks)

Module-2

- 3 a. With usual notations, derive an expression for temperature distribution for infinite fin. State the assumptions made. (10 Marks)
b. Find the amount of heat transfer through iron fin of thickness 5 mm, height 50 mm and width 100 cm. Take atmospheric temperature as 28°C and $K = 50 \text{ W/m-K}$. The $\text{HTC} = 10 \text{ W/m}^2\text{-K}$ the temperature difference at the base of the fin = 80°C . Estimate the efficiency of the fin. (10 Marks)

OR

- 4 a. With usual notations derive an expression for temperature distribution through a body for lumped parameter analysis in terms of Biot number and Fourier number. (10 Marks)
b. Mild Steel Sphere of 15 mm dia initially at 625°C is exposed to current of air at 25°C with $\text{HTC } h = 120 \text{ W/m}^2\text{-K}$. Calculate:
(i) Time required to cool the sphere to 100°C
(ii) Initial rate of cooling in $^\circ\text{C/sec}$.
(iii) Total energy removed for one minute. The thermophysical properties for MS are $K = 43 \text{ W/m-K}$, $C = 474 \text{ J/m}^3\text{-K}$, $\rho = 7850 \text{ kg/m}^3$ and $\alpha = 0.045 \text{ m}^2/\text{sec}$. (10 Marks)

Module-3

- 5 a. Write a note on spectral and total emissive power of a body. (08 Marks)
b. Write a short note on the concept of black body and grey body. (04 Marks)
c. The average solar radiation flux on the earth's atmosphere is 1353 W/m^2 . Calculate the temperature of sun (a black body) having diameter $1.392 \times 10^6 \text{ km}$ and has a mean distance of $1.496 \times 10^8 \text{ km}$ from the earth's atmosphere. State any assumption made. (08 Marks)

OR

- 6 a. Explain Wein's displacement law, Kirchoff's law and Max Plank's law. (10 Marks)
 b. Two large parallel planes with emissivity of 0.6 are at 900 K and 300 K. A radiation shield with one side polished and having emissivity of 0.05, while the emissivity of other side is 0.4 is proposed to be used. Which side of the shield to face the hotter plane, if the temperature of shield is to be kept minimum? Comment on your answer. (10 Marks)

Module-4

- 7 a. Explain the concept of development of boundary layer over a flat plate with different zones. (10 Marks)
 b. Atmospheric air at 2°C and free stream velocity of 20 m/s flows over 1.5 m long flat plate maintained at uniform temperature of 88°C. Calculate:
 (i) The average HTC 'h' over the region of laminar boundary layer.
 (ii) Average H.T.C. (Heat Transfer Coefficient) for entire length of plate 1.5 m.
 (iii) Total Heat Transfer Rate. Take critical Reynolds number $Re_c = 2 \times 10^5$. (10 Marks)

OR

- 8 a. Explain the significance of Reynolds number, Prandtl Number, Nusselt number and Grashof number with equations. (10 Marks)
 b. Calculate the total heat loss from a human body, assuming as vertical cylinder, 30 cm in dia and 175 cm in height stand in still air at 13°C. Take the skin temperature as 37°C and emissivity as 0.4. (10 Marks)

Module-5

- 9 a. Define heat exchanger and classify them. (04 Marks)
 b. Derive an expression for Log Mean Temperature Difference (LMTD) for counter flow heat exchanger. State the assumptions made. (08 Marks)
 c. A heat exchanger is required to cool 55000 kg/hr of alcohol from 66°C to 40°C using 44,000 kg/hr of water entering at 5°C. Calculate:
 (i) Exit temperature of water
 (ii) Heat transfer
 (iii) Surface area required for parallel flow and counter flow type heat exchanger design and comment on the results overall HTC $U = 580 \text{ W/m}^2\text{-K}$, $C_{p(\text{alcohol})} = 3760 \text{ J/kg-K}$, $C_{p(\text{water})} = 4180 \text{ J/kg-K}$. (08 Marks)

OR

- 10 a. Define film wise and drop wise condensation process. (04 Marks)
 b. With a neat sketch, explain the modes of pool boiling. (08 Marks)
 c. Steam at 0.065 bar condenses on a vertical plate 0.6 m square. If the surface temperature of the plate is maintained at 15°C, estimate the rate of condensate. The properties of condensate at mean temperature 26.4°C are, $\rho = 1000 \text{ kg/m}^3$, $\mu = 864 \times 10^{-6} \text{ N-S/m}^2$, $K = 0.913 \text{ W/m-K}$, $h_{fg(\text{latent heat})} = 2412 \times 10^3 \text{ J/kg-K}$. (08 Marks)

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18ME641

Sixth Semester B.E. Degree Examination, June/July 2023 Non - Traditional Machining

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define Non-traditional Machining. What are the need of NTM? Explain briefly. (08 Marks)
- b. What are difference between Conventional and Non – conventional Machining? (06 Marks)
- c. List and explain the various factors to be considered for selection of NTM process. (06 Marks)

OR

- 2 a. Give classification of NTM process. (08 Marks)
- b. List applications of NTM. (06 Marks)
- c. List any 3 advantages and limitations of NTM. (06 Marks)

Module-2

- 3 a. With neat sketch, explain USM process. (10 Marks)
- b. Explain with neat diagram, process parameters in USM. (10 Marks)

OR

- 4 a. Explain with neat sketch, working principle of Abrasive Jet Machining and also give advantages of AJM. (10 Marks)
- b. With the neat sketch, explain Water Jet Machining process and also give advantages and limitations of WJM. (10 Marks)

Module-3

- 5 a. With neat sketch, explain the working of ECM process. (10 Marks)
- b. With neat sketch, explain ECG. Also give the advantages and limitations of ECG. (10 Marks)

OR

- 6 a. Explain the following in Chemical Machining Process : (08 Marks)
 - i) Maskants
 - ii) Etchants.
- b. List out advantages and applications of Chemical Machining. (06 Marks)
- c. Write a short note on Chemical Blanking. (06 Marks)

Module-4

- 7 a. Explain with neat sketch the mechanism of metal removal in Electric discharge machining and also give applications. (10 Marks)
- b. Explain Die Electric Medium, its functions and desirable properties in EDM process. (10 Marks)

OR

- 8 a. With a neat sketch, explain Plasma Arc Machining Process. (10 Marks)
- b. Discuss some of the important considerations in the design of Plasma torch in PAM. (10 Marks)

Module-5

- 9 a. With a neat sketch, explain working principle of Laser Beam Machining. (08 Marks)
b. What are the advantages and disadvantages of LBM process? (06 Marks)
c. List the limitations and applications of LBM proces. (06 Marks)

OR

- 10 a. Explain with the help of neat diagram Principle of Electron Beam Machining (EBM). (10 Marks)
b. What are the advantages , disadvantages and applications of EBM process? (10 Marks)

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Sixth Semester B.E. Degree Examination, June/July 2023

Composite Materials Technology

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. What are composites? Explain the classification of composite based on reinforcement and matrix. (10 Marks)
- b. Define the following: i) Reinforcement ii) Matrix iii) Coupling agents
iv) Coatings v) Fillers. (10 Marks)

OR

- 2 a. List the different types of reinforcements. Explain any five. (10 Marks)
- b. Explain with a neat sketch different types of bonding at the interface. (10 Marks)

Module-2

- 3 a. Differentiate between thermoset and thermoplastic. (10 Marks)
- b. Explain the following methods of processing of PMC's:
i) Filament winding process
ii) Pultrusion process. (10 Marks)

OR

- 4 a. Explain with a neat sketch hand lay-up and autoclave bag moulding process. (10 Marks)
- b. Enumerate the applications of composite in the field of
i) Electronics and electricals
ii) Recreational and sports equipments
iii) Aerospace industries. (10 Marks)

Module-3

- 5 a. Explain the processing of CMC's using following methods:
i) Hot processing ii) Infiltration. (10 Marks)
- b. Explain self-propagation high temperature synthesis. (10 Marks)

OR

- 6 a. Explain with a flow chart processing of carbon-carbon composite. (10 Marks)
- b. Explain the oxide powder in the tube method for making high-temperature superconducting composite. (10 Marks)

Module-4

- 7 Explain the following:
a. Nanocomposite
b. Self-heating composite
c. Bio-composites
d. Hybrid composite. (20 Marks)

OR

- 8 a. Explain the following characterization methods of composites:
 i) Tensile ii) Compressive iii) Flexural iv) Inter laminar shear strength. (10 Marks)
 b. Explain the following impact properties: i) Izod ii) Charpy iii) Drop-weight. (10 Marks)

Module-5

- 9 a. Derive an expression for Young's modulus of FRP composite in iso-stress condition. (10 Marks)
 b. Consider a unidirectionally reinforced glass fiber/epoxy composite. The fibers are continuous and 60% by volume. The tensile strength of glass fiber is 1GPa and the Young's modulus is 70GPa. The tensile strength of epoxy matrix is 60MPa and its Young's modulus is 3GPa. Compute the Young's modulus and the tensile strength of the composite in the longitudinal direction. (10 Marks)

OR

- 10 a. Derive relationship between engineering constants and reduced stiffness and compliances. (10 Marks)
 b. For a unidirectional glass/epoxy lamina, the constituent material properties are as follows:
 $E_f = 76\text{GPa}$, $\gamma_f = 0.2$, $G_f = 35\text{GPa}$, $E_m = 3.6\text{GPa}$, $\gamma_m = 0.3$, $G_m = 1.4\text{GPa}$.
 Consider zero void content and fiber volume fraction of 0.6.
 i) Determine the composite longitudinal modulus, transverse modulus, major Poisson's ratio and in-plane shear modulus.
 ii) Apply a longitudinal force on the lamina and determine the ratio of axial forces shared by fibers and matrix. (10 Marks)

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Sixth Semester B.E. Degree Examination, June/July 2023

Non-Conventional Energy Sources

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Enlist and explain the merits and demerits of any three non-conventional Energy Sources. (10 Marks)
- b. Explain Tar Sands and Oil shale as energy sources and mention their limitations. (10 Marks)

OR

- 2 a. With schematic representation, explain mechanism of absorption, scattering beam and diffuse radiation received at earth's surface. (10 Marks)
- b. Explain with a neat sketch, explain the working of pyrenometer. (05 Marks)
- c. Explain briefly the need for alternate energy sources. (05 Marks)

Module-2

- 3 a. Define the following term with respect to solar radiation:
i) Hour angle ii) Declination angle iii) Zenith angle iv) Latitude angle
v) Solar Azimuth angle. (10 Marks)
- b. Calculate the day length of location (latitude $22^{\circ} 00' W$, $73^{\circ} 10' E$) during the month of March 1. (05 Marks)
- c. With the usual expression for flux explain beam and diffuse radiation on a tilted surface. (05 Marks)

OR

- 4 a. With a neat sketch explain working of liquid flat-plate collector. (08 Marks)
- b. Describe solar pond for solar energy collection and storage. (07 Marks)
- c. Explain how solar energy can be used for drying with a neat sketch. (05 Marks)

Module-3

- 5 a. List and discuss the various parameters that affect the performance of collector. (10 Marks)
- b. Explain the heat transfer process in LFPc with neat sketch and write the energy balance equation, explaining each terminal. (10 Marks)

OR

- 6 a. Explain the working principle and I-V characteristics of a solar PV cell. (10 Marks)
- b. Define : i) Collector efficiency factor ii) Collector heat removal factor of LPFc write the expression for the above. (05 Marks)
- c. What are the applications of solar PV cell? (05 Marks)

Module-4

- 7 a. Describe the main consideration in selecting the site for wind generators. (10 Marks)

- b. Wind blows with a velocity of 15 m/s at 15°C and 1 std. atm. pressure. The turbine diameter is 120m with operating speed of 40 rpm at maximum efficiency. Propeller type wind turbine is considered. Calculate the following :
- Total power density in the wind stream
 - Maximum obtainable power density
 - Obtainable power density
 - Total power
 - Torque at max η
 - Maximum axial thrust
- Assume $R = 0.287 \text{ kJ/kgK}$, $\eta = 35\%$. (10 Marks)

OR

- 8 a. Explain with a sketch, the closed Rankine cycle OTEC system. (10 Marks)
 b. Explain briefly the harnessing of Tidal energy. (05 Marks)
 c. Explain the advantages and disadvantages of Tidal energy. (05 Marks)

Module-5

- 9 a. State the environmental problem associated with geothermal energy conversion. (05 Marks)
 b. List the factors affecting biogas generation. (05 Marks)
 c. Sketch and explain the working of a fixed dome type biogas plant used in India. (10 Marks)

OR

- 10 a. What are the different methods of hydrogen production? Describe electrolytic method of hydrogen production. (10 Marks)
 b. Briefly explain the safe utilization of hydrogen energy. (05 Marks)
 c. Describe various methods of storage of hydrogen. (05 Marks)

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GBCS SCHEME

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18ME81

Eighth Semester B.E. Degree Examination, June/July 2023 Energy Engineering

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. What is pulverised coal? What are the advantages and limitations of pulverised coal? (10 Marks)
- b. Briefly explain the various steps involved in coal handling. (10 Marks)

OR

- 2 a. Explain the working principle of Benson boiler, with a neat sketch. (10 Marks)
- b. Explain common methods used for controlling super heat temperature of the steam. (10 Marks)

Module-2

- 3 a. Explain the working principle of pyranometer and pyrheliometer with a neat sketch. (10 Marks)
- b. With the help of a neat sketch, explain the extraction of solar energy from solar ponds. (10 Marks)

OR

- 4 a. Explain the working of floating drum biogas plant with a neat sketch. (10 Marks)
- b. Explain the working of updraft gasifier with a neat sketch. (10 Marks)

Module-3

- 5 a. With a neat sketch, explain the working of vapor dominated geothermal power plant. (10 Marks)
- b. With a neat sketch explain the harnessing tidal energy by the arrangement of double basin tidal power plant. (10 Marks)

OR

- 6 a. What are the properties of wind and explain the problems associated with the wind power. (10 Marks)
- b. With a neat sketch, explain Darrieus type wind machines and list the advantages and disadvantages. (10 Marks)

Module-4

- 7 a. With a neat sketch, explain medium and low head power plant (hydroelectric). (10 Marks)
- b. The mean monthly discharge for 12 months at a particular site of river is tabulated below:

Month	Discharge in millions of Cubic meter/month	Month	Discharge in millions of Cubic meter/month
May	500	October	2000
June	200	November	1500
March	1500	December	1500
July	2500	January	1000
August	3000	February	800
September	2400	March	600

- (i) Draw hydrograph and flow duration curve for the above and find average monthly flow.
- (ii) Determine the power available at mean flow of water if available head is 80 m at the site and overall efficiency of generation is 80%. Take 30 days in a month. **(10 Marks)**

OR

- 8** a. With a diagram, explain Open cycle or Claude cycle OTEC system. **(10 Marks)**
- b. With a diagram, explain Closed or Anderson OTEC system. **(10 Marks)**

Module-5

- 9** a. Explain the principle of radioactive decay, half life, fusion and fission in nuclear energy. **(10 Marks)**
- b. Explain with neat sketch of components of nuclear reactor. **(10 Marks)**

OR

- 10** a. Explain the working principle of pressurized water reactor with a neat sketch. **(10 Marks)**
- b. Explain the working principle of homogeneous graphite reactor and gas cooled reactor (indirect circuit gas cooled reactor) with a neat sketch. **(10 Marks)**

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CBCS SCHEME

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18ME824

Eighth Semester B.E. Degree Examination, June/July 2023

Automobile Engineering

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain with neat sketch the working of pump circulation type cooling system. (10 Marks)
b. Mention the necessity of valve clearance. Explain the working of overhead valve mechanism with a neat sketch. (10 Marks)

OR

- 2 a. Explain the different types of combustion chamber used in CI engines with suitable sketch. (10 Marks)
b. Explain with a neat diagram, pressure type lubricating system. (10 Marks)

Module-2

- 3 a. Explain with a neat sketch the working of a torque converter. (10 Marks)
b. Explain with a neat sketch the working of constant mesh type gear box. (10 Marks)

OR

- 4 a. Explain the working of airbrake system with a neat sketch. (10 Marks)
b. Explain Hotchkiss drive and Torque drive with a diagram. (10 Marks)

Module-3

- 5 a. Explain with a neat sketch the working of electronic ignition system. (10 Marks)
b. Explain with a neat diagram, the working of air suspension system. (10 Marks)

OR

- 6 a. Explain the working of power steering system with a neat diagram. (10 Marks)
b. Explain the working of a battery ignition system of a multicylinder engine with a diagram. (10 Marks)

Module-4

- 7 a. Mention the objectives of supercharging. Explain the different methods of supercharging. (10 Marks)
b. Explain alternate fuels for compression ignition engines. (10 Marks)

OR

- 8 a. Explain the working of common rail direct injection system with the help of a neat sketch. (10 Marks)
b. Explain the working of a turbo charging system with a neat block diagram. (10 Marks)

Module-5

- 9 a. Explain the working of a catalytic converter with the help of a neat sketch. (10 Marks)
b. Explain with a neat diagram, evaporator loss control system. (10 Marks)

OR

- 10 a. Explain how EGR controls emission of oxides of nitrogen with a neat sketch. (10 Marks)
b. Explain the different types of emission from IC engine and mention its effects. (10 Marks)

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Important Note : 1. On completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages.
2. Any revealing of identification, appeal to evaluator and /or equations written eg, 42+8 = 50, will be treated as malpractice.